



NCIE-006

Memory & Context Architecture Specification

**In development — for human review of surgical
amendment — Version 1.2**

Consolidates persistent Memory and Context architecture: classes, temporal integrity, provenance, authorization and Institutional Knowledge lifecycle

Model-independent: Memory survives ARGUS model/provider replacement and does not depend on any conversation-history feature

Traces to approved NCIE-002 v0.5 (architecture), approved NCIE-003 v0.4 (canonical data), approved NCIE-004 v1.2 (engineering), approved NCIE-005 v1.3 (ARGUS behavior) and approved NCIE-007 v1.1 (AI/Agent/Model orchestration)

Prepared: 04 September 2026

Maintained by: National Communications Intelligence Ecosystem (NCIE)

Document Governance

Document	NCIE-006 Memory & Context Architecture Specification
Edition	In development — Version 1.2 (VPF & Dynamic Agent Context Surgical Consistency Amendment)
Baseline authorized	NCIE-006 Skeletal v0.1 — approved for full expansion; 36-chapter structure locked
Requirements authority	NCIE-001 Master Product Requirements Document, Version 1.3 — APPROVED
Architecture authority	NCIE-002 System Architecture & Technical Design, Version 0.5 — APPROVED
Canonical data authority	NCIE-003 Data Model & Data Dictionary, Version 0.4 (Canonical Data Architecture Baseline) — APPROVED
Engineering baseline	NCIE-004 Technology Stack & Engineering Blueprint, Version 1.2 (Approved Engineering Baseline) — APPROVED
ARGUS behavioral authority	NCIE-005 ARGUS Intelligence Assistant Specification, Version 1.3 — APPROVED
Orchestration authority	NCIE-007 AI, Agent & Model Orchestration Specification, Version 1.1 — APPROVED
Prior baseline	NCIE-006 Memory & Context Architecture Specification v1.1 (IN DEVELOPMENT — FOR HUMAN REVIEW) — historically preserved
Owner	National Communications Intelligence Ecosystem (NCIE) Architecture Function
Reviewer	David King Boison (PhD) — Academic Intelligence Center
Status	IN DEVELOPMENT — FOR HUMAN REVIEW OF SURGICAL AMENDMENT until explicitly approved; the v1.1 approved-pending-review baseline remains historically preserved until v1.2 is reviewed and approved

0.1 Authority Chain

NCIE-001 requirements → NCIE-002 architecture → NCIE-003 canonical data → NCIE-004 engineering baseline → NCIE-005 ARGUS behavioral authority → NCIE-006 Memory & Context architecture. NCIE-006 implements and consolidates Memory/Context requirements established upstream; it does not reinterpret them, and it does not duplicate NCIE-007 (AI/Agent Orchestration), NCIE-009 (Security/IAM), NCIE-010 (Evidence/Provenance/Audit), NCIE-011 (API/Integration), NCIE-014 (Database & Storage Engineering) or NCIE-016 (Testing).

0.2 Token-Conservation Convention

Every chapter opens with a short Upstream block citing the specific upstream chapter and stating only the Memory/Context consequence — never a restatement of the upstream architecture. Major concepts (Memory, Context, Institutional Knowledge, Decision-Time, Knowledge Time, Reconciliation, Knowledge Candidate, Context Envelope, Memory Class) are defined authoritatively once, in the chapter that owns them, and referenced thereafter.

0.3 Governing Invariants Register

These 12 invariants bind every chapter of this document and are stated once here rather than re-derived repeatedly:

- ARGUS ≠ INSTITUTIONAL MEMORY.
- MEMORY ≠ CURRENT TRUTH.
- MEMORY ≠ EVIDENCE.
- MEMORY ≠ INSTITUTIONAL KNOWLEDGE.
- INSTITUTIONAL KNOWLEDGE ≠ RULE.
- CURRENT STATE ≠ HISTORICAL STATE ≠ DECISION-TIME STATE.
- SEMANTIC SIMILARITY ≠ AUTHORITY OR PRECEDENT.

- MODEL REPLACEMENT ≠ MEMORY RESET.
- PRIVATE CONTEXT ≠ SHARED CONTEXT.
- CONTEXT PROPAGATION ≠ PERMISSION PROPAGATION.
- REMEMBERED AUTHORIZATION ≠ CURRENT AUTHORIZATION.
- PRIOR HUMAN DECISION ≠ CURRENT ACTION AUTHORITY.

0.4 Human Review Focus Classification

Every chapter's Human Review Focus item carries one of six labels: RESOLVED BY UPSTREAM BASELINE; PROPOSED DESIGN DEFAULT; INSTITUTIONAL CONFIRMATION REQUIRED; SOURCE/WORKFLOW DISCOVERY REQUIRED; LEGAL/POLICY CONFIRMATION REQUIRED; or SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED — each additionally marked Blocking or Non-Blocking. Chapter 35 consolidates every open item, extracted directly from the chapters so the register cannot drift from its sources (mechanically verified per §35.4).

0.5 Status Convention

Skeletal-structure approval authorizes expansion; it does not approve any individual architectural interpretation or Proposed Design Default. Every chapter in this edition is stamped IN DEVELOPMENT — FOR HUMAN REVIEW and remains so until explicitly approved. A technology named from NCIE-004 retains its existing NCIE-004 approval status here — using it in an NCIE-006 example never promotes it.

0.6 Document Amendment History (v1.1 → v1.2)

Version: v1.1

Change: Initial full expansion from the approved NCIE-006 Skeletal v0.1 baseline. Status: IN DEVELOPMENT — FOR HUMAN REVIEW (historical baseline preserved unchanged above).

Version: v1.2

Change: VPF & Dynamic Agent Context Surgical Consistency Amendment (Chapters 1, 2, 3, 4, 5, 11, 14, 15, 17, 18, 22, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36). Implements the mandatory NCIE-003 v0.4 handover for Agent Task Context, Agent Run Context, VPF Artifact References, Promotion History and Context Expiry, and extends the 12 existing Governing Invariants with the minimum necessary Agent/VPF consistency extensions.

Authority: Approved NCIE-001 Master PRD v1.3, approved NCIE-002 System Architecture & Technical Design v0.5, approved NCIE-003 Data Model & Data Dictionary v0.4, approved NCIE-004 Technology Stack & Engineering Blueprint v1.2, approved NCIE-005 ARGUS Intelligence Assistant Specification v1.3, and approved NCIE-007 AI, Agent & Model Orchestration Specification v1.1.

Scope: Controlled surgical consistency amendment — not a Memory redesign, new Knowledge architecture, Agent orchestration rewrite, IAM rewrite, database design or vector-database procurement exercise. The 36-chapter structure is unchanged. Chapters 6-10, 12, 13, 16, 19, 20, 21, 23, 24 were reviewed for Dynamic Agent/VPF contradiction and required no substantive change beyond what Chapter 5's consolidated review already states applies to the Memory-class family.

Human Review Register: 16 → 20 items (4 new: HR6-4-2, HR6-18-2, HR6-22-1, HR6-33-2; 0 removed). Blocking 13 → 17; Non-Blocking unchanged at 3.

Non-Impact: No change to the 12 Governing Invariants, Memory Class boundaries, Institutional Knowledge lifecycle, Evidence/Rule governance, or existing Technology Registry status of any NCIE-004-named technology.

Status: v1.2 IN DEVELOPMENT — FOR HUMAN REVIEW OF SURGICAL AMENDMENT. The v1.1 baseline remains historically preserved until v1.2 is reviewed and approved.

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Memory & Context Mandate, Authority Chain & Specification Boundary

Establish NCIE-006 as the detailed Memory/Context authority without redefining upstream architecture.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.20 §20.2, §20.12 — Memory tiers and the formal promotion boundary are architecturally fixed; NCIE-006 details their implementation. | NCIE-003 Ch.22 — Memory Object, Knowledge Candidate, Institutional Knowledge entities are canonical; NCIE-006 specifies the services and lifecycle around them.

1.1 Mandate

NCIE-006 consolidates persistent Memory and Context architecture into an implementable specification. It preserves model independence, Human-Primary governance, temporal integrity, provenance and authorization, and does not redesign the upstream architecture that already fixes these requirements.

1.2 Specification Boundary — What NCIE-006 Does Not Own

Table 1. NCIE-006 specification boundary.

Concern	Owned By
AI/agent/model orchestration engineering	NCIE-007; NCIE-006 specifies only what Memory/Context must expose to it
Identity, authorization enforcement engineering	NCIE-004 Ch.23; NCIE-009; NCIE-006 specifies authorization requirements, not IAM implementation
Evidence object lifecycle, chain-of-custody engineering	NCIE-003 Ch.17; NCIE-010; NCIE-006 references Evidence, never redefines it
API/event transport engineering	NCIE-004 Ch.8-9; NCIE-011; NCIE-006 specifies service contracts, not wire protocols
Physical database/search/graph/vector technology selection	NCIE-004 (Technology Decision Matrix); NCIE-014; NCIE-006 specifies logical storage roles only
Comprehensive test-suite implementation	NCIE-016; NCIE-006 Ch.34 specifies required test categories, not the full verification programme
ARGUS conversational/analytical behavior	NCIE-005; NCIE-006 specifies what Memory ARGUS may retrieve, not how ARGUS behaves

1.3 Structural Change Control

The 36-chapter structure is locked. A material contradiction or omission discovered during expansion is flagged STRUCTURAL CHANGE REQUIRED — HUMAN REVIEW rather than silently resolved. None has been raised in this edition.

1.4 Current Authority Chain & Agent/VPF Boundary Extension (v1.2)

Current authorities: approved NCIE-004 v1.2 (engineering), approved NCIE-005 v1.3 (ARGUS behavior), approved NCIE-007 v1.1 (AI/Agent/Model orchestration) — do not treat NCIE-001 v1.2, NCIE-002 v0.4, NCIE-003 v0.3, NCIE-004 v1.1 or NCIE-005 v1.2 as current authority merely because this document's v1.1 front matter cited them. NCIE-006 additionally specifies: what Context an Agent may receive, how Agent Context is bounded, how Agent/VPF provenance is referenced, how Context expires, how historical Agent/VPF state participates in Decision-Time reconstruction, and how dependent Memory/Knowledge changes trigger impact/revalidation obligations. It does not specify Agent Factory or Runtime implementation (NCIE-004 v1.2), or orchestration doctrine (NCIE-007 v1.1).

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm the acceptance authority for NCIE-006 as a whole and for individual chapter sign-off.

Traceability: NCIE-002 Ch.20; NCIE-003 Ch.22.

Memory Doctrine, Invariants & Epistemic Boundaries

Lock Memory ≠ Evidence, Memory ≠ Current Truth, Memory ≠ Institutional Knowledge, Knowledge ≠ Rule, and ARGUS ≠ Memory.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.20 §20.2 — ARGUS is not institutional memory; Memory ≠ Current Truth is architecturally fixed. | NCIE-003 Ch.22 §22.2 — Institutional Knowledge ≠ Rule; promotion requires recorded human review.

2.1 Authoritative Definitions

Table 2. Core term definitions, stated once and referenced throughout this document.

Term	Definition
Memory	A persistent record of something previously observed, decided, discussed or concluded. It is retrospective by nature — it records what was, not what currently is.
Institutional Knowledge	A Memory-derived generalisation that has been deliberately reviewed and approved through the lifecycle in Ch.18-19, and is applicable within a defined scope (Ch.19 §19).
Evidence	An authoritative artifact with source, acquisition and validation lineage (NCIE-003 Ch.17) — governed entirely outside NCIE-006; Memory may reference Evidence, never substitute for it.
Rule	A governed, effective-dated parameter or threshold (NCIE-003 Ch.24) — governed entirely outside NCIE-006; Institutional Knowledge may motivate a Rule Change Candidate but never directly modifies a Rule.
Agent Context	A governed task/run-scoped assembly of authorized Context required for an Agent's execution (Ch.4, Ch.14). It is not a new authority source, a new institutional Memory tier, a permission container, or permanent Agent-owned Memory.
Agent Task Context	The Context justified by an Agent's Task Contract/purpose (Ch.4 §4.2) — the intended scope, distinct from what is actually delivered (Agent Run Context).
Agent Run Context	The exact Context actually assembled/provided for a particular Agent Run (Ch.4 §4.2, Ch.14) — attributable to that Run for provenance and Decision-Time reconstruction.

2.2 Epistemic Boundaries

- ARGUS ≠ Institutional Memory — ARGUS has no memory store of its own; every recollection is retrieved from governed NCIE-006 services (front matter §0.3).
- Memory ≠ Current Truth — a Memory Object is never presented as live authoritative state without reconciliation (Ch.13).
- Memory ≠ Evidence — a Memory Object may reference Evidence; it never becomes Evidence by being remembered.
- Memory ≠ Institutional Knowledge — a raw Memory Object requires deliberate promotion (Ch.18) before it carries institutional authority.
- Institutional Knowledge ≠ Rule — approved Knowledge informs human judgment and may motivate a Rule Change Candidate; it never directly alters a governed Rule (NCIE-003 Ch.24).

2.3 Model Independence

Model Replacement ≠ Memory Reset. Replacing the ARGUS model or provider (NCIE-004 Ch.18) never erases or resets Memory or Institutional Knowledge, since both live in governed NCIE-006 services entirely outside the model/provider boundary. Chapter 34's acceptance tests demonstrate this directly.

2.4 Agent/VPF Epistemic Extensions (v1.2)

Extending §2.2's five epistemic boundaries: AGENT RUNTIME STATE ≠ INSTITUTIONAL MEMORY — temporary Agent execution state (intermediate task state, scratch state, transient tool results) requires the applicable existing Memory lifecycle before it becomes governed Memory. AGENT CONTEXT ≠ AGENT AUTHORITY and AGENT CONTEXT PROPAGATION ≠ PERMISSION PROPAGATION — the presence of data in a Context Envelope never itself authorizes disclosure, Tool execution, model use, cross-context retrieval or consequential action. EPHEMERAL AGENT EXPIRY ≠ REQUIRED PROVENANCE DELETION. VALIDATED AI RESULT ≠ INSTITUTIONAL KNOWLEDGE and VPF PASS ≠ KNOWLEDGE PROMOTION (Ch.18). VPF VALIDATION STATE ≠ HUMAN APPROVAL STATE. HISTORICAL AGENT / VPF STATE ≠ CURRENT ELIGIBILITY (Ch.11-13). RECOVERY ≠ REAUTHORIZATION (Ch.32). SEMANTIC RETRIEVAL ≠ AUTHORITY TO DISCLOSE (Ch.17, Ch.23-24). AI ANALYSIS OF EVIDENCE ≠ EVIDENCE ITSELF — an Agent/ARGUS summary of Evidence remains AI-derived and never replaces the underlying Evidence identity/provenance governed entirely outside NCIE-006 (§2.1's Evidence definition). This edition also implements the mandatory NCIE-003 v0.4 handover to NCIE-006 for Agent Task Context, Agent Run Context, VPF Artifact References, Promotion History and Context Expiry — these are surgical additions to this chapter's scope, not a new Memory doctrine.

Human Review Focus

- **[RESOLVED BY UPSTREAM BASELINE]** All five epistemic boundaries in §2.2 are fixed by NCIE-002 Ch.20 and NCIE-003 Ch.22; no institutional decision is required to adopt them.

Traceability: NCIE-002 Ch.20 §20.2, §20.12; NCIE-003 Ch.22.

Memory Class Taxonomy

Define Conversation, Preference, Workspace, Room, Investigation, Situation, Historical, Decision and Institutional Knowledge memory.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.20 §20.2 — Private/Shared/Institutional tiers are the architectural basis this taxonomy details into nine operational classes.

3.1 Memory Class Architecture

Figure 1. Memory Class Architecture — authority flows only through deliberate promotion (Ch.18); no Memory class silently escalates its own authority.

SOURCE	RELATIONSHIP	TARGET
Conversation Memory / User Preference Memory	carry no institutional authority into	Workspace, Room, Investigation, Situation Memory
Workspace, Room, Investigation, Situation Memory	may be finalized (never auto-converted) into	Decision Memory
Decision Memory / Historical Memory	may be generalised through deliberate promotion (Ch.18) into	Institutional Knowledge
Institutional Knowledge	never demotes back to, nor is derived automatically from,	Conversation or Preference Memory

3.2 Memory Class Matrix

Nine Memory classes are distinguished throughout this document and never collapsed into one generic store or lifecycle:

Table 3. Memory Class Matrix — nine distinguished classes, each independently governed.

Memory Class	Purpose	Authority	Owner Chapter	Reconciliation Required?
Conversation Memory	Low-authority conversational continuity	None — never overrides current state, Evidence, Rules, Findings, Decisions or authorization	Ch.5	No
User Preference Memory	Low-risk personalization	None — never alters permissions, Evidence meaning, Rules, analytical truth or Decision authority	Ch.6	No
Workspace/Session Continuity Memory	Resumable individual work state	Low — reflects a user's own prior working state	Ch.7	Yes, on resume
Collaborative Room Memory	Structured multi-user analytical reasoning	Working analysis — Hypothesis/Consensus, never Finding/Decision	Ch.8	Yes, on re-entry
Investigation Memory	Investigative evidence relationships, hypotheses, tasks	Working analysis	Ch.9	Yes, on resume
Situation Memory	Evolving cross-domain analytical state	Working analysis	Ch.10	Yes, on resume
Decision Memory	High-integrity Decision-Time snapshots	High — protected historical record of institutional action	Ch.11	N/A — immutable once finalized
Historical Memory	Bitemporal record of past canonical state	Reflects what was known/true at a point in time	Ch.12	Yes, before consequential reuse
Institutional Knowledge	Reviewed, approved, applicability-scoped generalisation	Institutional — requires human approval (Ch.18-19)	Ch.18-19	Yes, via challenge/review (Ch.20)

3.3 Class-Specific Attributes

Each class in §3.2 additionally has its own Scope, Retention, Authorization and Retrieval behavior, detailed in its owning chapter — this matrix is the index, not the full specification for any one class.

3.3 Agent Runtime State Classification (v1.2)

Dynamic Agent execution does not create a new permanent institutional Memory class. Temporary state required during Agent execution is classified as Task/Run Context (Ch.4) or Ephemeral Runtime State — not as a new 'Agent Memory' tier — and persistence into governed Memory follows the applicable existing Memory-class lifecycle already defined in this chapter.

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm retention, access or ownership rules that differ by Memory class beyond what each owning chapter proposes.

Traceability: NCIE-002 Ch.20 §20.2.

Context Taxonomy & Context Envelope

Define purpose-bounded, authorized, temporal, provenance-aware context passed among users, services, tools and ARGUS.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.19 §19.6 — Conversation/Tool Invocation/Proactive Contribution objects carry resolved context; NCIE-006 formalizes the envelope structure.

4.1 Context Envelope Specification

The Context Envelope is the single implementable structure passed among users, services, tools and ARGUS. Fields are included only where they carry real weight — not merely because they are convenient:

Table 4. Context Envelope field specification.

Field	Purpose
Context ID	Stable reference for this specific context instance (correlation/audit).
User	The human identity the context is resolved for (never a machine-inferred identity, NCIE-005 Ch.4 §4.2).
Session / Workspace / Room / Investigation / Situation / Task	Which of these scopes the context is bound to — mutually clarifying, not all populated simultaneously.
Operator / Domain / Geography	The NCIE-003 canonical scope the context concerns.
Time Mode	Current / Historical / Decision-Time (Ch.12-13) — never left implicit.
Time Range / Decision-Time Cutoff	The specific bound applicable when Time Mode is not Current.
Purpose	Why this context is being assembled (feeds Ch.14's minimum-necessary retrieval).
Classification	The governing classification tier (NCIE-003 Ch.31) of the context's most sensitive component.
Authorization Context	The effective authorization the context was assembled under (Ch.25) — never expanded after assembly.
Provenance	Where each component of the context originated (Ch.27).
Correlation ID	Cross-service tracing reference (NCIE-004 Ch.9).

4.2 Context Taxonomy

- Private Context — visible only to the originating user (Ch.16).
- Shared Context — visible to authorized participants of a specific Room/Investigation/Situation (Ch.16).
- Institutional Context — promoted, broadly applicable Institutional Knowledge (Ch.16, Ch.18-19).

4.4 Agent Context Envelope Extension (v1.2)

The existing Context Envelope (§4.1) is extended to bind, where applicable: Agent Run Reference, current Authorization Context, and the Agent Task Context / Agent Run Context distinction (Ch.2 §2.1). CONTEXT ENVELOPE ≠ AUTHORIZATION GRANT — the Envelope records/propagates applicable authorized Context; it does not create new authority. Agent Context must support MINIMUM NECESSARY AGENT CONTEXT (synthesis/execution does not imply access to everything available to ARGUS or NCIE) and remains bound to its Task Contract, purpose, Agent Run and current authorization — Context assembled for one purpose does not become a general reusable permission. Task-scoped Agent Context supports expiry consistent with the Agent/task lifecycle; expiry does not delete required Audit, Provenance, canonical Agent Run History, VPF History or Legal-Hold material. After an Ephemeral Agent Run completes/expires, runtime access to task Context ends per policy while historical references needed for audit/Decision-Time remain governed separately. No new wire-format schema is invented here; NCIE-011 owns API detail.

Human Review Focus

- **[PROPOSED DESIGN DEFAULT]** The Context Envelope field set in §4.1 is proposed as complete for launch; additional fields may be added additively without a breaking change (NCIE-003 Ch.2 §2.8 pattern).

Traceability: NCIE-002 Ch. 19 §19.6.

Conversation Memory

Specify low-authority conversational continuity, retention, summaries and invalidation.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-005 Ch.5 — Conversational assistance grounds answers in authoritative tool results; Conversation Memory supplies continuity only, never authority.

5.1 Conversation Memory Boundary

Conversation Memory preserves continuity across turns and sessions. It never overrides Current Authoritative Domain State, Evidence, Rules, Findings, Decisions or authorization. A user previously saying something does not make it current institutional truth — Conversation Memory is retrieved and labelled per NCIE-005 Ch.11's epistemic-type matrix ("Historical Memory"), never presented as a live fact.

5.2 Lifecycle

Table 5. Conversation Memory lifecycle.

State	Trigger	Resulting State
Active	Turn recorded during a live session	Retrievable as recent continuity
Aged	Retention policy threshold reached (Ch.28)	Retrievable as Historical Memory only, not default continuity
Invalidated	User/administrator requests removal, or content found materially inaccurate	Retained per Ch.28 §28 rules (INVALIDATED ≠ DELETED), excluded from active retrieval

5.3 Agent Participation Boundary (v1.2)

Agent participation in a conversation does not automatically create permanent Agent-owned Conversation Memory; persisted conversational state remains governed by this chapter's existing architecture. This extends to Chapters 6-10 (Preference, Workspace/Session, Room, Investigation, Situation Memory), reviewed together for Dynamic Agent consistency: PREFERENCE ≠ PERMISSION (a preference such as 'always use Agents' cannot override Context/authorization policy, Ch.6); Workspace/Session state from one Agent Run does not become unrestricted cross-session Context (Ch.7); ROOM MEMBERSHIP / TASK PARTICIPATION ≠ CROSS-ROOM ACCESS, and Agent participation in a Room does not make it a permanent Room member (Ch.8); Investigation Agent Context remains bounded to the applicable Investigation/purpose/authorization, and semantic retrieval must not bridge unrelated Investigations merely because content is similar (Ch.9); cross-domain Situation Memory synthesis does not create cross-domain permission (Ch.10). Isolation among Users, Rooms, Investigations, Workspaces, Cases/Situations and Agent Runs is preserved throughout — Dynamic Synthesis must not weaken existing boundaries.

Human Review Focus

- **[LEGAL/POLICY CONFIRMATION REQUIRED]** Confirm Conversation Memory retention period — no universal period is assumed; classified per Ch.28's class-specific retention requirement.

Traceability: NCIE-005 Ch.5, Ch.11.

User Preference Memory

Define permitted low-risk preferences without allowing preferences to change authority or truth.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-005 Ch.28 — Personalization is permitted for presentation/notification/voice/verbosity/workspace; preferences never alter authorization or institutional truth.

6.1 Preference Memory Boundary

Table 6. User Preference Memory boundary (mirrors NCIE-005 Ch.28 §28.2, restated here as the canonical Memory-side rule).

Preference May Influence	Preference Shall Never Alter
Presentation, language, voice, notification preferences, accessibility settings	Permissions
Low-risk personalization generally	Evidence meaning
—	Regulatory Rules
—	Analytical truth
—	Decision authority

Traceability: NCIE-005 Ch.28.

Workspace & Session Continuity Memory

Preserve resumable individual work state independently of runtime sessions.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.3 §3.1 — Workspace layout/context state must survive across sessions independently of runtime session objects.

7.1 Workspace Continuity

Workspace Memory preserves an individual's resumable work state — layout, active filters, open panels, last context — independently of any specific runtime session, browser tab or ARGUS conversation instance. A session ending does not destroy the workspace state a user would expect to resume.

7.2 Reconciliation on Resume

On resume, Workspace Memory is reconciled against current authoritative state (Ch.13) before being presented — a saved filter referencing a since-closed Investigation, for example, surfaces as CHANGED or SUPERSEDED rather than silently reopening stale context.

Traceability: NCIE-002 Ch.3 §3.1.

Collaborative Room Memory

Persist structured multi-user reasoning independently of the raw transcript.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.20 §20.5 — Room Memory persists structured reasoning objects, distinct from the raw transcript, per the Collaborative Reasoning State architecture. | NCIE-003 Ch.21 §21.3 — Room, Thread, Question, Hypothesis, Assumption, Contradiction, Evidence Gap, Task, Human Judgment, Decision Point, ARGUS Intervention, Consensus, Dissent and Room Snapshot are canonical persistent objects.

8.1 Room Memory Is Not a Transcript

Room Memory persists the structured analytical objects named above as first-class records — the raw message transcript may supplement this structure but never substitutes for it. A 'summarize this room' request is answered from the structured objects, not by re-reading the transcript from scratch.

8.2 Voice-Derived Room Memory

VOICE-DERIVED MEMORY. Where verbal Collaborative Intelligence is used, Raw Audio, Speech-to-Text Transcript, Structured Room Memory and Formal Record are distinct objects, never automatically the same thing. A transcript error never silently alters a persisted Hypothesis, Human Judgment, Decision Point or authorization. Where a material spoken contribution is mistranscribed, correction preserves the original transcript state where required, the corrected transcript, correcting actor, correction time, and the affected structured objects.

8.3 Room Snapshot and Cross-Session Continuity

A Room Snapshot captures the Collaborative Reasoning State at a point in time, enabling 'where were we?' reconstruction by combining the snapshot with current authoritative state (never presenting a stale snapshot as current without reconciliation, Ch.13).

Human Review Focus

- **[SOURCE/WORKFLOW DISCOVERY REQUIRED]** Confirm audio retention policy specifics with NCA (inherits NCIE-005's open item on audio/transcript retention as a separate decision, per §0.2's no-duplication rule — see NCIE-005 Ch.29 §29 review).

Traceability: NCIE-002 Ch.20 §20.5, §20.10; NCIE-003 Ch.21.

Investigation Memory

Preserve investigative evidence relationships, hypotheses, contradictions, gaps, tasks and human judgments.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.15 §15.3 — Investigation stage progression (Detection→Suspicion→Investigation→Validation→Blocking→Outcome) is canonical; Investigation Memory persists its working analytical state.

9.1 Investigation Memory Behavior

Investigation Memory preserves evidence relationships, hypotheses, contradictions, evidence gaps, tasks and human judgments accumulated during an investigation, supporting pause/resume across sessions, shifts, users, ARGUS model changes and system restarts — resumption never depends on the original chat/session being available.

9.2 Sensitivity Inheritance

Investigation Memory tied to Anti-Fraud/SIMBOX work inherits sensitive-tier classification (NCIE-003 Ch.15, Ch.31) automatically; it is never treated as ordinary working-analysis memory merely because it uses the same underlying structure as Ch.8's Room Memory.

Traceability: NCIE-003 Ch. 15.

Situation Memory

Preserve evolving Situation context, timelines and cross-domain analytical state.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.18 — Situation is a canonical entity distinct from Condition/Alert/Investigation/Case; Situation Memory persists its evolving cross-domain analytical state.

10.1 Situation Memory Behavior

Situation Memory preserves timelines and cross-domain analytical state across the Situation's lifecycle (NCIE-003 Ch.18), supporting the same pause/resume guarantees as Investigation Memory (Ch.9) — across sessions, shifts, users, model changes and restarts.

Traceability: NCIE-003 Ch. 18.

Decision Memory & Decision-Time Context

Preserve high-integrity snapshots of what was known when consequential human Decisions were made.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.16 §16.3, Ch.25 §25.5 — Decision-Time Snapshot and Decision-Time Context are canonical; Decision Memory is their high-integrity persisted form.

11.1 Decision Memory Contents

Decision Memory preserves, where applicable:

- Decision ID, Decision Authority, Decision Time.
- Decision-Time Snapshot (the bounded fact/evidence/rule-version set available at that timestamp).
- Evidence Available at Decision Time; Applicable Rule Versions.
- Known Hypotheses, Known Contradictions, Known Evidence Gaps at that time.
- Human Rationale; ARGUS Contribution (distinctly attributed, NCIE-005 Ch.27).
- Subsequent Review Relationship (appeal/reversal, linked forward, never rewriting the original).

11.2 Decision-Time Definition

Decision-Time is formally defined here and referenced, not redefined, elsewhere in this document: the specific timestamp as of which a Decision's supporting context is bound. Reconstruction implements AS KNOWN AT T, never WHAT WE KNOW NOW ABOUT T — the system supports both views but never confuses them (Ch.12 covers the general bitemporal mechanism this relies on).

11.3 Integrity Protection

Table 7. Decision Memory integrity requirements.

Requirement	Behavior
Immutability	A finalized Decision-Time Snapshot is protected against ordinary modification once created
Subsequent events as new objects	Correction/review/appeal/outcome are linked as subsequent objects/events, never rewrites of the original snapshot
Sensitive-content access	Access to a Decision record does not automatically grant access to every underlying restricted Evidence item (Ch.25); redacted/limited reconstruction is supported
Outcome ≠ quality/causation	A later-known outcome informs learning; it never automatically rewrites the historical assessment of whether the Decision was reasonable given what was known then

11.4 Prior Decision Is Never Automatic Current Authority

PRIOR HUMAN DECISION ≠ CURRENT ACTION AUTHORITY (front matter §0.3). A finalized Decision recorded in Decision Memory may inform current human reasoning — as precedent, context or pattern — but it is never automatically replayed as authorization for a new blocking, tracking, regulatory or other consequential action. A new consequential action requires its own current human Decision Authority and current Decision-Time context (§11.1-§11.2); ARGUS may surface a prior Decision as relevant context (NCIE-005 Ch.24) but never initiates or authorizes a new action on the strength of the prior Decision alone.

11.3 Agent/VPF Decision Memory and Decision-Time Extension (v1.2)

Where an institutional Decision relied materially on AI/Agent output, Decision Memory may reference the applicable Agent Run, Agent Result Package, VPF Validation Artifact and Human Decision/Approval Record as distinct, non-equated objects — AI / AGENT / VPF HISTORY ≠ HUMAN DECISION AUTHORITY: Decision Memory records what informed/occurred; it does not transfer authority to the Agent or VPF. Historical Decision-Time reconstruction (Ch.12) must be able to identify the actual Agent Definition Version, Agent Run, Task Contract, Context Available,

Model Route, Tool Outcomes, Agent Result and VPF State available at the historical cutoff where materially relevant. Later Agent Registration, Promotion, Evaluation, Restriction or Retirement — and later VPF revalidation/invalidation — must not be used to rewrite what was true at the cutoff; later state may be shown separately, never substituted in. For present consequential use, Current-State Reconciliation (Ch.13) resolves historical Memory against current Authorization, Agent Eligibility, Model Eligibility, Tool Eligibility, VPF Profile/Validator State and external-system state where applicable: HISTORICAL AGENT / MODEL / TOOL / VPF ELIGIBILITY ≠ CURRENT ELIGIBILITY.

Traceability: NCIE-003 Ch.16, Ch.25, Ch.26.

Historical Memory & Bitemporal Reconstruction

Implement occurrence/effective time and knowledge/record time for defensible historical reconstruction.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.5 — Occurrence/effective time and knowledge/record time are the canonical bitemporal axes; this chapter is their Memory-service realisation.

12.1 Bitemporal Definitions

Table 8. Bitemporal time axes (formally defined here; NCIE-003 Ch.5 establishes the platform-wide principle this chapter applies to Memory specifically).

Time Axis	Definition
Occurrence / Valid Time	When the remembered fact actually happened or was true.
Knowledge / Record Time	When NCIE came to know/record the fact — never substituted with ingestion time where they genuinely differ.

12.2 Historical Reconstruction Flow

Figure 2. Decision-Time / historical reconstruction flow — later evidence never contaminates the reconstructed view.

01	Reconstruction request: 'what was known as of knowledge-time T?'
02	Bind to canonical facts with knowledge time ≤ T (NCIE-003 Ch.25 pattern)
03	Bind to Rule Versions effective at T (NCIE-003 Ch.24)
04	Bind to Evidence registered with knowledge time ≤ T
05	Exclude any fact/rule/evidence with knowledge time > T
06	Return AS KNOWN AT T, with later developments available separately as SUBSEQUENTLY LEARNED

12.3 Correction Without Overwrite

Historical knowledge state is never overwritten merely because current understanding changes. A correction to historical Memory uses an explicit CORRECTION, AMENDMENT, NEW VERSION, INVALIDATION or SUPERSESSSION relationship (Ch.20 detail for Knowledge specifically) — the institution's prior understanding remains reconstructable even after it is known to have been wrong.

Human Review Focus

- [PROPOSED DESIGN DEFAULT]** UTC as the canonical knowledge-time storage basis is proposed, consistent with NCIE-003 Ch.5 §5.8's platform-wide default.

Traceability: NCIE-003 Ch.5, Ch.25.

Current-State Reconciliation

Reconcile remembered state against authoritative current domain state before consequential reuse.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.20 §20.10 — Previous collaborative state must be reconciled with current authoritative NCIE state — the architectural mandate this chapter operationalizes for all Memory classes.

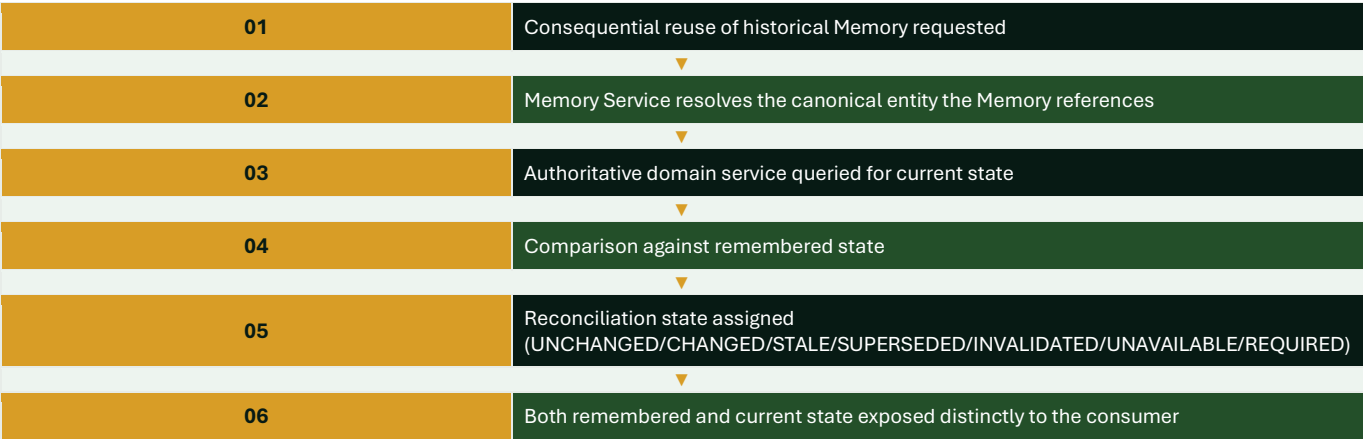
13.1 Reconciliation States

Table 9. Current-State Reconciliation states — historical Memory is never silently updated; history and current state are always separately exposed.

State	Meaning
UNCHANGED	Current authoritative state matches remembered state exactly.
CHANGED	Current state differs from remembered state in a way that does not invalidate the memory's historical accuracy.
STALE	Remembered state has exceeded its freshness expectation and has not been re-verified.
SUPERSEDED	A newer canonical version exists; the remembered version is no longer current.
INVALIDATED	The remembered fact has been formally invalidated at its source.
CURRENT STATE UNAVAILABLE	The authoritative source cannot be reached to verify (Ch.30 §30 failure behavior).
RECONCILIATION REQUIRED	A material discrepancy exists that requires explicit resolution before consequential reuse.

13.2 Current-State Reconciliation Flow

Figure 3. Current-State Reconciliation flow.



13.3 Priority Rule

When a user requests current operational state, authoritative domain services take precedence over remembered historical state. Memory may provide context; it never replaces authoritative current-state retrieval (NCIE-005 Ch.23 restates this at the ARGUS behavior layer; this chapter is its Memory-service-level enforcement point).

Traceability: NCIE-002 Ch.20 §20.10; NCIE-005 Ch.23.

Context Assembly & Minimum-Necessary Retrieval

Assemble only authorized, relevant and purpose-bounded context for humans, ARGUS and tools.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.24 §24.3 — Context Builder assembles only authorization-filtered, minimal context per request — this chapter is its Memory-side implementation.

14.1 Context Assembly Flow

Figure 4. Context Assembly Flow — minimum-necessary retrieval, never a full institutional context dump.

01	Requesting service/ARGUS states Purpose and Context Envelope (Ch.4)
02	Authorization filter applied (Ch.25) — never expanded beyond requester's effective scope
03	Classification filter applied — sensitive-tier content requires explicit eligibility
04	Temporal Mode applied (Current/Historical/Decision-Time, Ch.11-13)
05	Relevance and Applicability filters applied (Ch.19 §19 scope matching for Institutional Knowledge)
06	Context Budget enforced — excess content trimmed by materiality, not truncated arbitrarily
07	Assembled Context Envelope returned

14.2 Assembly Considerations

- Purpose — what the context is for shapes what is retrieved.
- Authorization and Classification — never assembled beyond the requester's effective scope.
- Temporal Mode — Current/Historical/Decision-Time are never mixed silently.
- Relevance and Applicability — Institutional Knowledge outside its applicable scope (Ch.19) is excluded, not force-included.
- Data Minimization and Context Budget — prioritised by authorization, purpose, authority, temporal applicability, materiality, evidence/contradiction relevance; never by semantic similarity alone.

14.3 ARGUS Does Not Access Raw Memory Stores

Figure 5. ARGUS's memory access path — no AI-only unrestricted retrieval path exists.

01	ARGUS
02	Context/Memory Retrieval Service
03	Authorization + Purpose + Temporal Filters
04	Memory/Knowledge Sources

14.2 Dynamic Agent Context Assembly (v1.2 — principal amendment)

Context Assembly is extended to support governed assembly for ARGUS Direct AI, Registered Agent Runs, Ephemeral Agent Runs and, where required, VPF Validation. Assembly considers the applicable Task Contract, Effective Principal, Purpose, Agent Definition, current Authorization, Memory/Evidence/Knowledge Scope, Temporal Cutoff, Classification and Resource/Context Limits. The assembled Agent Context must be attributable to the exact

Agent Run — a generic retrieved document set is not sufficient provenance. CONTEXT SELECTION ≠ TASK AUTHORIZATION: Assembly filters what may be provided; it does not itself authorize the underlying task/action. AGENT DEFINITION ≠ CONTEXT AUTHORITY and TASK CONTRACT ≠ CONTEXT AUTHORITY apply identically. If required authorization/scope cannot be established, Assembly does not default to broader Context — it uses the applicable safe failure/review state.

Traceability: NCIE-002 Ch.24 §24.3.

Context Propagation Across Services & Workflows

Propagate canonical context without propagating permissions or corrupting temporal meaning.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.4 §4.4 — Zero-trust authorization is re-evaluated at every service boundary; context propagation never substitutes for it.

15.1 Context Propagation ≠ Authorization Propagation

A user navigating Dashboard → Situation → Investigation → Evidence → ARGUS may retain relevant context (operator, geography, active thread). Navigation never expands access — every destination independently applies its own authorization check (Ch.25), regardless of what context was carried forward.

15.2 Permission-Change Propagation

- If a user's authorization changes during an active session, Room or Investigation, subsequently assembled context reflects the new authorization immediately — authorization is never assumed valid for the lifetime of a session.
- Revocation after sensitive context has already been retrieved is handled via: future retrieval block; UI redaction/removal where technically feasible; cache invalidation; ARGUS context removal; search/graph access restriction; and audit. Already-perceived human information cannot be technically "unseen" — this is an acknowledged limit, not a false guarantee.

15.2 Agent and Multi-Agent Propagation (v1.2)

Propagation extends to Agent execution while preserving CONTEXT PROPAGATION ≠ PERMISSION PROPAGATION. Where bounded Agent-to-Agent task delegation is permitted upstream (NCIE-007 v1.1), child Context is constrained by the intersection of the effective principal's authorization, the parent's delegated scope, the child task, the child Agent Definition, and current Context policy — the parent's full Context Envelope is never copied automatically; only what the child task requires and is authorized to receive propagates. Lineage sufficient to identify Parent Run, Child Run, Delegated Task, Context Sources and Context Reduction/Transformation is preserved where material. Context propagation must not provide an Ephemeral Agent with Agent Factory authority: RECURSIVE DYNAMIC AGENT SYNTHESIS REMAINS DISABLED by the current upstream default.

Traceability: NCIE-002 Ch.4 §4.4.

Private, Shared & Institutional Context Boundaries

Govern movement from private context to Room-shared context and ultimately to reviewed institutional knowledge.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.20 §20.9 — Shared room context never pools participant permissions; Private User Memory is not automatically shareable.

16.1 Private → Shared → Institutional Context Transition

Figure 6. Private → Shared → Institutional context transition — each step is a deliberate, human-governed action, never automatic.

01	Private Context (single user, e.g. Preference/Conversation Memory, Ch.5-6)
02	Deliberate sharing action into a Room/Investigation/Situation (Ch.8-10)
03	Shared Context — visible only to that Room/Investigation/Situation's authorized participants
04	Knowledge Candidate proposed from Shared Context pattern (Ch.18)
05	Human review and approval (Ch.18-19)
06	Institutional Context — Approved Institutional Knowledge, broadly applicable within its scope

16.2 Shared Room Disclosure

A Collaborative Room's shared disclosure is calculated per-participant from governed authorization, never as a permission union: one participant's ability to access an object never means ARGUS may expose it to every Room participant (NCIE-002 Ch.20 §20.9). ARGUS's effective Room context is restricted to information appropriately shareable in that context — private context from one participant never silently becomes ARGUS-shared Room context.

Traceability: NCIE-002 Ch.20 §20.9; NCIE-003 Ch.4 §4.72-4.79.

ARGUS Memory Retrieval Interface

Define governed Memory Service interfaces consumed by ARGUS without giving ARGUS unrestricted institutional memory access.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-005 Ch.23 — ARGUS retrieves memory via governed services and reconciles it against current state; NCIE-006 specifies the interface those retrievals actually use.

17.1 Governed Retrieval Interface

ARGUS consumes Memory exclusively through the Memory/Context Retrieval Service (Ch.14's assembly flow), never a direct store connection. Every retrieval carries the requesting identity's effective authorization, purpose, and temporal mode; none of these may be widened by the fact that the requester is ARGUS rather than a human.

17.2 Retrieval Result Labels

Returned Memory/Knowledge visibly distinguishes its category — never collapsed into one generic "memory result":

- CURRENT AUTHORITATIVE CONTEXT
- HISTORICAL MEMORY
- DECISION-TIME CONTEXT
- APPROVED INSTITUTIONAL KNOWLEDGE
- PRIOR HUMAN JUDGMENT
- ARGUS-GENERATED HISTORICAL ANALYSIS
- UNVALIDATED KNOWLEDGE CANDIDATE

These labels are the Memory-service-side source of NCIE-005 Ch.11's ARGUS Output/Epistemic-Type Matrix — that chapter specifies how ARGUS presents them; this chapter specifies that the Memory Service produces them accurately in the first place.

17.3 Agent Retrieval and Model-Invocation Provenance (v1.2)

Agent retrieval through this interface remains subject to Authorization, Purpose, Classification, Context Scope, Temporal Scope and Provenance exactly as ARGUS retrieval is. RETRIEVABLE ≠ AUTHORIZED TO DISCLOSE applies particularly to semantic/vector retrieval (Ch.23-24). Retrieval results preserve enough source identity to support Evidence Citation, Knowledge Provenance, Decision-Time, VPF and Audit; where retrieval includes AI-generated artifacts, their AI Origin, Agent/ARGUS Origin, VPF State, Qualification and Human Modification State are preserved where applicable, and such artifacts are never surfaced as source Evidence. Where Context is injected at model invocation, minimization applies identically and sufficient provenance is retained for reconstruction — this chapter does not duplicate NCIE-004/007's Model Gateway or prompt-construction specification.

Traceability: NCIE-005 Ch.11, Ch.23.

Institutional Knowledge Candidate Lifecycle

Define how lessons and reusable insights become governed Knowledge Candidates.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.22 §22.4 — Promotion to Institutional Knowledge requires a recorded human Review State; a Knowledge Candidate never self-promotes.

18.1 Knowledge Candidate Definition

A Knowledge Candidate is a proposed generalisation derived from Memory, awaiting human review before it may carry institutional authority (Ch.2 §2.1's authoritative definition applies).

18.2 Institutional Knowledge Promotion Lifecycle

Figure 7. Institutional Knowledge Promotion Lifecycle — the main forward path; §18.3's table below covers the full state machine including challenge/supersession/retraction branches.

01	CANDIDATE (proposed, origin traceable)
02	REVIEW (human evaluation, separation of duties where policy requires)
03	APPROVED (applicability scope defined, Ch.19)
04	ACTIVE (in force within scope)

18.3 Full Lifecycle State Machine

Table 10. Institutional Knowledge Candidate lifecycle state machine.

State	Trigger	Permitted Actor	Resulting State
CANDIDATE	Proposed from Human Analyst, Collaborative Room, Investigation, Situation, Post-Incident Review, Expert Contribution or ARGUS Proposal	Any of the above; origin remains traceable after promotion	REVIEW
REVIEW	Submitted for human evaluation	Authorized reviewer, distinct from proposer where policy requires (Ch.20 §20.5)	APPROVED or REJECTED
APPROVED	Reviewer approves with defined applicability scope (Ch.19)	Authorized approver	ACTIVE
ACTIVE	In force within its applicability scope	N/A (state, not action)	CHALLENGED, REVIEW_DUE, SUPERSEDED, RETRACTED or EXPIRED
CHALLENGED	Material conflict identified (Ch.20)	Any authorized user/ARGUS may flag; only a human resolves	ACTIVE (challenge resolved) or REVIEW (re-review triggered)
REVIEW_DUE / SUPERSEDED / RETRACTED / EXPIRED	Review date reached / newer version approved / formally withdrawn / applicability period lapsed	Per Ch.19 policy	Terminal or REVIEW, per Ch.19 §19

18.4 ARGUS Role

ARGUS may propose a Knowledge Candidate, organize supporting evidence, surface contradictions, and prepare review material. ARGUS does not approve — this remains true even where the candidate originated from ARGUS itself (NCIE-005 Ch.24 §24.1 restates this at the behavioral layer).

18.4 Validated AI Result and Agent Result Knowledge Boundary (v1.2)

VALIDATED AI RESULT ≠ INSTITUTIONAL KNOWLEDGE: VPF validation establishes the applicable validation state of an AI artifact; it does not promote that artifact into Institutional Knowledge. VPF PASS ≠ KNOWLEDGE PROMOTION — Knowledge promotion remains governed by this chapter's existing Human-Primary lifecycle. AGENT RESULT ≠ KNOWLEDGE CANDIDATE BY DEFAULT: where upstream governance permits creating a Knowledge Candidate from an Agent Result, that is an explicit governed transition, not an automatic one. Where an Agent/Validated AI Result contributes to a Knowledge Candidate, Source Evidence, AI Artifact, Agent Run, VPF Validation and Human Promotion/Review are preserved as distinct provenance elements. Where a human materially modifies an AI-generated or VPF-validated artifact before Memory persistence or Knowledge promotion, the original and human-modified artifacts remain distinct provenance states — a materially modified artifact does not automatically inherit the original VPF validation state, and applicable revalidation/governance is required before it is represented as currently validated. Where used Knowledge is retracted or superseded, dependent AI output is not continued to be presented as current without the applicable impact assessment/revalidation.

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm whether separation of duties between Candidate Contributor, Reviewer and Approver is institutionally required, or whether one authorized role may perform more than one step.

Traceability: NCIE-003 Ch.22 §22.4; NCIE-005 Ch.24.

Institutional Knowledge Versioning, Applicability & Review

Define scope, effective period, limitations, review, challenge, supersession and retraction.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.22 §22.3 — Applicability Scope and Knowledge Version are canonical Institutional Knowledge attributes.

19.1 Applicability Scope

Approved Institutional Knowledge identifies its applicable Domain, Operator, Geography, Technology, Time, Methodology, Situation Type, Regulatory Context or other relevant boundary. Contextual lessons are never silently universalized — if retrieved Knowledge appears used outside its approved scope, ARGUS may warn KNOWLEDGE APPLICABILITY REVIEW REQUIRED (NCIE-005 Ch.24 restates this as ARGUS behavior); it never silently extends the scope itself.

19.2 Review-Due Behavior

Table 11. Institutional Knowledge review and ownership-transfer behavior.

Situation	Behavior
Review date reached	Knowledge class/policy determines whether it remains ACTIVE — REVIEW DUE or moves to SUSPENDED PENDING REVIEW; where policy is unknown, classified INSTITUTIONAL CONFIRMATION REQUIRED, never assumed universally
Ownership changes (staff departure, reorganization, role/department change)	Original Authorship, Review, Approval and Contribution Provenance are preserved; changing the owner never rewrites history
No accountable current owner	Enters GOVERNANCE REVIEW REQUIRED rather than remaining indefinitely ownerless

19.3 Versioning

Each Knowledge Version retains its own effective period and content; retrieval of current guidance always resolves to the active version, while historical reconstruction (Ch.12) can recover which version was in force at a prior knowledge time.

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm review-due default behavior (ACTIVE-REVIEW-DUE vs SUSPENDED) per Knowledge class where NCA policy is not otherwise specified.

Traceability: NCIE-003 Ch.22 §22.3.

Knowledge Challenge, Contradiction & Impact Analysis

Handle conflicts between current evidence and approved knowledge through explicit review and downstream impact analysis.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-005 Ch.24 — ARGUS may identify a contradiction between current Evidence and approved Knowledge; humans govern the Knowledge response.

20.1 Knowledge Challenge, Supersession & Retraction Lifecycle

Figure 8. Knowledge Challenge / Supersession / Retraction lifecycle.

01	Material conflict identified between current Evidence and Approved Knowledge (human or ARGUS)
02	KNOWLEDGE REVIEW REQUIRED state generated — never silently resolved by picking one side
03	Human review of the conflict
04	Outcome: Knowledge reaffirmed, superseded by a new version, or retracted
05	SUPERSEDED/RETRACTED recorded — historical traceability retained (SUPERSEDED ≠ DELETED; RETRACTED ≠ NEVER EXISTED)
06	Current retrieval stops presenting superseded/retracted Knowledge as active guidance

20.2 Retraction Impact Analysis

If Knowledge is retracted, active Rooms, Investigations, Situations, Briefings, Reports or Decision-Support Packages that materially relied upon it are flagged for review — historical artifacts are never silently rewritten as a result (Ch.22 formalizes the dependency-tracking mechanism this relies on).

20.3 Re-Review Triggers

Review date reached, material contradiction, underlying Rule change, definition change, methodology change, major incident, repeated challenge, or scope-expansion request may each trigger review — none of them automatically retracts Knowledge; they trigger the human review process in Ch.18 §18.2.

Traceability: NCIE-005 Ch.24; NCIE-003 Ch.22.

Knowledge Graph Context & Memory Relationships

Use canonical graph relationships for memory/knowledge without converting connectivity into fact, cause or culpability.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.19 §19.2 — Graph Relationship ≠ Causality is the platform-wide invariant; this chapter applies it to Memory/Knowledge relationships specifically.

21.1 Memory Authority Hierarchy

Figure 9. Memory Authority Hierarchy — the chain of deliberate promotion from Evidence to Institutional Knowledge.

SOURCE	RELATIONSHIP	TARGET
Evidence (NCIE-003 Ch.17)	may be cited by	Memory Object
Memory Object	may be generalised into	Knowledge Candidate (Ch.18)
Knowledge Candidate	requires human Review State to become	Approved Institutional Knowledge
Approved Institutional Knowledge	may motivate (never directly modify)	Rule Change Candidate (NCIE-003 Ch.24)
Memory / Knowledge Graph edges	are correlations, never	Causality or Culpability claims (NCIE-003 Ch.19 §19.2)

21.2 Graph Authorization

Graph traversal authorizes Node, Edge, Connected Object and, where relevant, Topology Disclosure. A hidden node must not be inferable merely through degree count, neighbor count, path existence or graph visualization — the same authorization discipline NCIE-003 Ch.19 §19.7 requires for the Intelligence Graph applies identically to the Memory/Knowledge Graph.

Traceability: NCIE-003 Ch.19 §19.2, §19.7.

Dependency Graph & Downstream Impact Tracking

Track typed dependencies so corrections and invalidations expose affected downstream artifacts.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.25 §25.4 — Correction Relationships trigger forward impact analysis via Dependency Edges; this chapter applies that mechanism to Memory/Knowledge.

22.1 Knowledge Graph / Dependency Relationship Architecture

Figure 10. Knowledge Graph / Dependency Relationship Architecture — typed dependencies used for impact tracking, not causal assertion (Ch.21 §21.1 restates the authority-hierarchy view of this same graph).

SOURCE	RELATIONSHIP	TARGET
Source	supports	Evidence
Evidence	supports	Memory / Hypothesis
Hypothesis	may inform (never automatically becomes)	Finding
Finding	supports	Decision
Decision / Memory	may generalise to	Institutional Knowledge
Institutional Knowledge / Decision	may be cited by	Report / Briefing

22.2 Typed Dependencies and Impact Analysis

If upstream Evidence changes, dependent objects identified via §22.1's typed edges are flagged for review — this never automatically reverses a human Decision (NCIE-003 Ch.25 §25.4's Impact-Review State pattern applies identically here).

22.3 Pending-Output Invalidation

If ARGUS has generated a draft briefing or analytical artifact from information subsequently invalidated before human approval, the artifact is flagged SOURCE STATE CHANGED — REVIEW REQUIRED rather than silently remaining apparently current.

22.4 Agent/VPF Dependency and Invalidation Architecture (v1.2)

The dependency graph is extended to material changes in Evidence, Knowledge, Agent Definitions, Models, Tools, Context and VPF Profiles, supporting identification of dependent Agent/AI artifacts for applicable impact analysis, revalidation, re-evaluation, qualification, invalidation or Human Review. DEPENDENCY ≠ AUTHORITY: identifying a dependency does not itself authorize any action on the dependent artifact. There is no automatic universal invalidation — a VPF Profile change, Agent Definition change, Model change, Tool change or Human Review change triggers targeted impact analysis on genuinely dependent artifacts, not a blanket invalidation of unrelated Memory. Historical Validation Artifacts are preserved regardless of downstream dependency changes.

Traceability: NCIE-003 Ch.25 §25.4.

Memory Search, Retrieval Modes & Ranking

Define Current, Historical and Decision-Time search with authority- and applicability-aware ranking.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.4 §4.89-4.93 — Search enforces authorization before returning results, including facets, snippets, autocomplete and semantic matches.

23.1 Search & Retrieval Architecture

Figure 11. Memory Search & Retrieval Architecture — mode-differentiated, authorization-filtered throughout.

01	Search Request (Current / Historical / Decision-Time mode)
02	Authorization Filter
03	Mode-Specific Query (canonical store / bitemporal store / decision-time-bound store)
04	Ranking (Ch.23 §23.2)
05	Labelled, Authorized Results (Ch.17 §17.2 result labels)

23.2 Ranking Factors

Ranking considers Authority, Applicability, Temporal Validity, Governance State, Relevance and Source Quality — never solely Recency or Semantic Similarity, which are inputs to relevance but never the entire ranking basis.

23.3 Retrieval Explainability

For material retrieval, authorized users can determine why an object was returned, along the dimensions in §23.2 plus Source and Retrieval Mode — without exposing implementation-sensitive scoring internals unnecessarily.

23.4 Search-Result Leakage Prevention

Authorization applies before disclosure through titles, snippets, counts, graph neighbors, semantic matches, autocomplete or facets — none of these surfaces is exempt from the same authorization check applied to the underlying object (NCIE-003 Ch.4 §4.89-4.93).

Traceability: NCIE-003 Ch.4 §4.89-4.93.

Semantic Memory & Vector Retrieval Boundaries

Use embeddings for discovery without making vector representations Evidence or Institutional Knowledge.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.26 §26.3 — Sensitive semantic/vector indexing requires isolation; NCIE-004 Ch.16 defines the classification-aware vector retrieval stack this chapter governs the Memory-side rules for.

24.1 Embeddings Are Derived, Not Authoritative

Embeddings are derived retrieval artifacts. They are not Evidence, Institutional Knowledge or canonical truth. Changing the embedding model never changes the substantive meaning or status of the underlying object — a re-embedding event (NCIE-004 Ch.16) is a retrieval-technology change, not a Memory/Knowledge change.

24.2 Sensitive Semantic Indexing

Highly sensitive information — Protected Identity, raw Ghana Card/Passport values, precise Anti-Fraud location, restricted Investigation Evidence, sensitive infrastructure data — is never placed into general-purpose vector infrastructure merely because semantic retrieval is useful. Where semantic retrieval over such content is required, it uses isolated, authorization-aware infrastructure per NCIE-004 Ch.16's index-isolation-by-classification-tier requirement.

Traceability: NCIE-002 Ch.26 §26.3; NCIE-004 Ch.16.

Memory Security, Authorization & Zero-Trust Controls

Apply object-, purpose-, classification- and temporal-aware authorization to all memory classes.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.4 — Zero-trust, four-layer authorization (role/object/field/action) applies to every Memory class independently.

25.1 Memory Authority Matrix

Table 12. Memory Authority Matrix — object-, purpose-, classification- and temporal-aware; nothing is transitively assumed.

Access Granted	Does NOT Automatically Grant
Access to current canonical objects	Historical access, Private Memory access, Room Memory access, Investigation Memory access, or Decision Memory access
Room membership	Access to every Evidence object referenced within the Room (NCIE-003 Ch.4 §4.69)
Access to a Decision record	Access to every underlying restricted Evidence item cited by it (Ch.11 §11.3)

25.2 Concurrency and Conflict

Concurrent human/system activity (two reviewers editing a Knowledge Candidate, a Room Snapshot during active collaboration, retraction while ARGUS retrieves the same object, permission revocation mid-session, correction during historical reconstruction) is never resolved by silent last-write-wins for consequential institutional state. A VERSION CONFLICT state requires reload, merge or human resolution according to object type.

25.3 Remembered Authorization Is Never Current Authorization

REMEMBERED AUTHORIZATION ≠ CURRENT AUTHORIZATION (front matter §0.3). A Memory object may record that a person held a role, permission or access grant at some past point — including within a Decision-Time Snapshot (Ch.11) or a Historical Memory reconstruction (Ch.12). That recorded fact is never itself treated as authorizing a new consequential action today. Every consequential action is authorized from the current governed IAM/authorization context (NCIE-004 Ch.23; NCIE-009), re-checked at the time of the new action, regardless of what Memory shows the person was permitted to do previously — a revoked, expired or role-changed permission is never reinstated by historical recollection. Current-State Reconciliation (Ch.13) applies to remembered authorization exactly as it applies to any other remembered state.

25.4 Zero-Trust Memory Access for Dynamic Agents (v1.2)

Zero-Trust Memory Access is extended to Dynamic Agents, distinguishing Human Principal, ARGUS, Agent Runtime Identity, Service Identity and VPF/Validator Identity where applicable. Agent Memory access remains bound to the effective-principal/purpose context while using a separate workload identity: DELEGATED TASK ≠ HUMAN CREDENTIAL DELEGATION — human credentials are never copied into Agent Context. AGENT IDENTITY ≠ AUTHORITY. Authorization is checked against the current authoritative source, not cached at Run start; mid-Run revocation applies to subsequent operations. Least privilege, purpose binding and access audit/denial apply identically to Agent access as to human access. CONTEXT EXFILTRATION BOUNDARY: reading Context does not authorize exporting, speaking, Tool transmission, persistence elsewhere, or disclosure to another Agent — each downstream use remains independently governed, including Tool-bound, model-bound, Agent-to-Agent, VPF and Human-Review-facing disclosure. SYSTEM ADMINISTRATION ≠ CONTENT AUTHORITY: administrative/operational access to Memory infrastructure is distinct from institutional authority to read substantive content, including for break-glass access.

25.5 Context Security and Isolation (v1.2)

Runtime isolation, context-at-rest and context-in-transit protections, and temporary Context storage controls apply to Agent Context exactly as to other governed Memory. Context caching does not create or extend authority: a cached Context is subject to the same current-authorization reconciliation as fresh assembly before reuse. Context integrity controls guard against poisoning/contamination from untrusted retrieved content. Sensitive Context is

minimized by default, and Context destruction (on expiry, revocation, or policy) follows this chapter's existing secure-disposition controls (Ch.28).

Traceability: NCIE-002 Ch.4.

NCIE-006 — Chapter 26 Expansion

Sensitive Memory, Protected Identity & Voice Context

Govern Protected Identity, Anti-Fraud location, voice transcripts/audio and other heightened-sensitivity context.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.14 §14.6, Ch.36 §36.3 — Ghana Card/Passport and Anti-Fraud location protections apply to Memory exactly as to canonical data. | NCIE-005 Ch.29 §29.1 — The Voice Interaction & Spoken-Disclosure Matrix governs what may be spoken; this chapter governs what is remembered about it.

26.1 Governed Sensitive Classes

- Protected Identity (Ghana Card, Passport) — masked/tokenized in Memory exactly as in canonical stores (NCIE-003 Ch.14 §14.6); never stored in plaintext in a Memory Object merely because it appeared in a conversation.
- Anti-Fraud location — inherits sensitive-tier isolation and heightened authorization wherever it appears in Memory (Ch.9 §9.2).
- Restricted Investigation Evidence and sensitive infrastructure data — classification inherited from the referenced canonical object, never downgraded by being remembered.

VOICE-DERIVED MEMORY. Voice/Speaker Recognition ≠ Authenticated Identity — Memory never records an inferred speaker identity as authoritative; identity comes from the governed NCIE authentication context (NCIE-005 Ch.4 §4.2), never from voice characteristics. Display Authorization ≠ Spoken-Disclosure Authorization — remembered authorization to view sensitive information never becomes permission to speak it aloud later; every synthesis still passes through NCIE-005 Ch.29's spoken-disclosure check at the time of speaking. Spoken-notification Memory: the fact that ARGUS spoke a notification is never treated as proof a human heard, understood, acknowledged or acted on it — acknowledgement, where it matters, is a separately stored governed event (Ch.11-style integrity, not inferred from audio).

26.2 Voice Object Distinctness

Raw Audio, Speech-to-Text Transcript, Structured Room/Investigation Memory and Formal Record are distinct objects (Ch.8 §8.2 for the Room-specific case). Audio retention and transcript retention are separate decisions — raw audio storage is never assumed merely because transcription occurred.

26.3 Voice Context Boundary for Dynamic Agents (v1.2)

VOICE CONTEXT ≠ AUTHENTICATED IDENTITY: speaker recognition, transcription or voice-derived Context alone does not establish current authorization, and does not create Agent authority. VPF PASS ≠ SPOKEN-DISCLOSURE AUTHORIZATION — retrievability of a sensitive artifact from Memory does not create permission to speak it; the same voice-sensitive-retrieval bounding (authorization, purpose, classification, Room/Investigation/User scope, temporal mode) applies to Agent-initiated retrieval as to ARGUS direct retrieval. Voice transcript provenance is preserved sufficiently for traceability and Decision-Time reconstruction without storing private chain-of-thought.

26.4 Protected Identity in Agent Context (v1.2)

Current authorization, purpose, classification, minimum-necessary scope and isolation apply to protected identity information reaching Agent Context exactly as elsewhere in this chapter. Dynamic Agent/VPF capability must not broaden protected-identity access, cross-user identity leakage exposure, or derived-sensitive-identity inference beyond what this chapter already governs. VPF and Human Review processes that reference protected identity remain subject to the same minimization and pseudonymization controls as any other consumer.

Human Review Focus

- [LEGAL/POLICY CONFIRMATION REQUIRED]** Confirm audio/transcript retention periods and applicable privacy policy — this is the same open question already raised at NCIE-005 Ch.29 §29-review; not duplicated here as a new decision.
- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED]** Confirm cross-border processing boundaries for any voice/transcript content that could leave an approved jurisdiction via cloud STT/TTS processing (NCIE-004 Ch.18's provider eligibility applies).

Traceability: NCIE-003 Ch. 14 §14.6, Ch.36; NCIE-005 Ch.29.

Memory Provenance, Attribution & Audit

Preserve human, ARGUS, workflow and source origins plus material retrieval and lifecycle audit.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-005 Ch.27 — AI origin is preserved permanently; human modification never erases it; no private chain-of-thought is stored.

27.1 Provenance Requirements

Material Memory/Knowledge objects preserve:

- Origin and Actor Type (human, ARGUS, workflow, source — always distinguishable).
- Source References; Created Time; Knowledge Time.
- Version; Classification; Review State.
- Correction/Supersession Relationships.

27.2 No Chain-of-Thought Requirement

Memory is never designed to store private model chain-of-thought. Defensibility comes from persisted structured artifacts — Questions, Hypotheses, Assumptions, Contradictions, Evidence Gaps, Evidence References, Tool Results, ARGUS Contributions and Human Responses — matching NCIE-005 Ch.27 §27.2 exactly, restated here as a Memory-architecture constraint rather than an ARGUS behavioral one.

27.3 AI-Generated Summaries

An ARGUS-generated Memory summary retains ARGUS Origin, Source References, Generation Time, applicable temporal cutoff, and Validation/Review State where required. Human adoption never erases AI provenance. A dependent summary whose underlying Memory materially changes supports STALE/REFRESH_REQUIRED rather than remaining silently presented as current.

27.4 Historical Prompt-Injection Boundary

Stored Conversation, Document, Evidence, Knowledge Candidate or Historical Note content may contain malicious instructions. Such content remains data — it never becomes executable ARGUS instruction merely because it is retrieved from Memory (NCIE-002 Ch.24 §24.4's ingestion principle applies identically to retrieval from Memory).

27.3 Agent/VPF Provenance and Audit (v1.2)

An authorized reviewer must be able to reconstruct the material relationship Human/System Request → Context Assembly → Agent Run → Evidence/Knowledge Dependencies → Model/Tool Provenance → Agent Result → VPF Validation → ARGUS Presentation → Human Decision/Modification, without requiring private chain-of-thought. Agent Context audit, VPF provenance audit and human-modification audit are preserved with the same discipline as this chapter's existing provenance/audit architecture: TELEMETRY ≠ AUDIT ≠ PROVENANCE — the three may correlate but are not interchangeable records.

Traceability: NCIE-002 Ch.24 §24.4; NCIE-005 Ch.27.

Retention, Legal Hold, Archival & Secure Disposition

Define differentiated retention, hold, archive and deletion behavior including derived artifacts.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.31 — Classification/privacy/retention taxonomy is canonical; this chapter applies class-specific retention to each of Chapter 3's nine Memory classes.

28.1 Class-Specific Retention

No universal Memory retention period exists. Each Memory class (Ch.3 §3.2) has its own retention policy hook; where the institutional/legal period is unknown, it is classified LEGAL/POLICY CONFIRMATION REQUIRED, never invented.

28.2 Holds Override Disposition

LEGAL HOLD, REGULATORY HOLD, INVESTIGATION HOLD and RECORDS HOLD override ordinary disposition for any Memory object they apply to, exactly as NCIE-003 Ch.31 §31.5 requires for canonical data generally.

28.3 Derived Deletion

Where an object is lawfully destroyed, its Search Index entry, Vector Embedding, Cache entry, Graph Materialization and Snippet are addressed explicitly — primary data is never deleted while retrievable derived copies remain unintentionally (NCIE-003 Ch.31 §31.4's Deletion Propagation mechanism applies to every derived Memory representation).

28.4 Personal Memory Deletion Boundary

Deletion of eligible personal Conversation or Preference Memory never silently destroys independent Formal Records, Decision Evidence, Room Records, Investigation Records or Audit Records that have a separate lawful institutional basis.

28.4 Ephemeral Agent Retention and Disposition (v1.2)

EPHEMERAL AGENT ≠ EPHEMERAL PROVENANCE: runtime expiry does not erase required Agent Definition, Agent Run, Task Contract, Result, VPF, audit or Decision-Time records. Runtime scratch-state cleanup follows this chapter's existing disposition controls, distinct from the retention of Agent Context, Validation Artifacts, Qualification History, Promotion History and Engineering Change Candidate history, all of which follow their own applicable retention/legal-hold rules. DELETION ≠ HISTORICAL FALSIFICATION: where deletion is required, deletion propagation must not leave a record implying an event never occurred where a tombstone/redaction is the governed alternative.

Human Review Focus

- **[LEGAL/POLICY CONFIRMATION REQUIRED]** Confirm retention periods per Memory class (Ch.3 §3.2) with NCA Legal/Records Management; none is assumed.
- **[LEGAL/POLICY CONFIRMATION REQUIRED]** Confirm how personal deletion rights interact with institutional records obligations where they may conflict — no universal precedence is assumed.

Traceability: NCIE-003 Ch.31.

Memory Storage Architecture & Canonical/Derived Stores

Define transactional, object, analytical, graph, search/vector and immutable storage roles without creating alternate truth.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-004 Ch.11-17 — Transactional/analytical/object/search/graph/vector/cache store roles are engineering-governed; this chapter assigns Memory's logical role to each without naming a product.

29.1 Logical Memory Storage Architecture

Figure 12. Logical Memory Storage Architecture — canonical vs derived roles; no alternate truth exists in a derived store.

SOURCE	RELATIONSHIP	TARGET
Transactional Memory Metadata	canonical for	Memory Object identity, state, provenance
Object/Document Store	canonical for	Room transcripts, briefing artifacts, large content bodies
Historical/Analytical Store	derived, rebuildable, canonical for	bitemporal historical query (Ch.12)
Knowledge Graph	derived, canonical for	typed Memory/Knowledge relationships (Ch.21)
Search Index / Vector Index	derived, rebuildable, never independently canonical	discovery and semantic retrieval (Ch.23-24)
Snapshot/Immutable Archive	canonical, append-only for	Decision-Time Snapshots and Room Snapshots (Ch.11, Ch.8)
Cache	derived, ephemeral accelerator over	all of the above

29.2 Technology Neutrality

This chapter specifies logical roles only. Physical technology selection remains governed by NCIE-004's Technology Decision Matrix; a technology named there retains its existing approval status when referenced here — this document never promotes a Proposed Design Default to Approved merely by using it in an example (NCIE-004 Ch.33 §33.1A's invariant applies identically).

29.3 Agent/VPF Canonical References (v1.2)

LOGICAL ENTITY ≠ SEPARATE DATABASE: Agent Definition, Agent Run, Task Contract, Agent Result Package, VPF Profile and VPF Validation Artifact references are logical Memory-layer concerns; this chapter's existing approved persistence architecture is reused where governance and performance requirements can be met, consistent with NCIE-004 v1.2's Registry review (Ch.33 §33.4). NCIE-006 references these canonical objects; it does not own their canonical storage, which remains NCIE-003 v0.4 (semantics) and NCIE-004 v1.2 / NCIE-014 (physical implementation).

Traceability: NCIE-004 Ch. 11-17, Ch.33.

Memory APIs, Events & Service Contracts

Define stable retrieval, reconciliation, historical-state and knowledge-lifecycle interfaces and events.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-004 Ch.8-9 — API/event contract engineering is governed upstream; this chapter defines the logical capability set those contracts must expose.

30.1 Service Capabilities

- GetCurrentContext, GetHistoricalContext, GetDecisionTimeContext — the three retrieval modes (Ch.23) as distinct calls, never one undifferentiated "get memory" endpoint.
- SearchMemory — authorization- and mode-aware (Ch.23).
- ReconcileMemory — invokes Ch.13's reconciliation flow explicitly.
- CreateKnowledgeCandidate, ReviewKnowledgeCandidate, ChallengeKnowledge, SupersedeKnowledge, RetractKnowledge — one call per lifecycle transition (Ch.18, Ch.20), never one generic "update knowledge" call.

Names may be normalized during downstream implementation (NCIE-011); the capability boundaries above are what must not be collapsed.

30.2 Material Events

- MemoryCreated, MemoryCorrected, MemoryInvalidated.
- KnowledgeCandidateCreated, KnowledgeApproved, KnowledgeChallenged, KnowledgeSuperseded, KnowledgeRetracted.
- DecisionSnapshotCreated.
- ReconciliationRequired.

Event payloads reference sensitive objects rather than copying them unnecessarily (NCIE-003 Ch.28 §28.2's data-minimization pattern applies identically to Memory events).

30.3 Agent/VPF Event Propagation (v1.2)

Material Agent/VPF lifecycle events use this chapter's existing event architecture for lifecycle/impact propagation. **EVENT ≠ CURRENT AUTHORITATIVE STATE:** events may trigger reconciliation/impact processing but do not replace current Registry/authorization/VPF sources, and do not themselves establish institutional authority. This applies identically to event replay, ordering and duplicate-event handling.

Traceability: NCIE-004 Ch.8-9; NCIE-003 Ch.28.

Scalability, Tiering, Performance & Observability

Define Hot/Warm/Cold tiering, retrieval SLOs, index health, reconciliation monitoring and governance metrics.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-004 Ch.31, Ch.35 — Hot/Warm/Cold tiering and performance targets are NCIE-004's proposed defaults; this chapter applies them to Memory specifically.

31.1 Tiering and Eventual Consistency

Derived representations (Search, Graph, Vector, Cache, Analytical) may update asynchronously. Each derived representation exposes whether it is CURRENT, LAGGING, STALE or UNAVAILABLE — a consequential current-state decision never depends unknowingly on stale derived state (Ch.13's reconciliation states are the canonical-store analogue of this derived-store concept).

31.2 Canonical-State Fallback

Where a derived store is stale or unavailable, the operation may fall back to the canonical store where technically feasible. Where fallback is not feasible, the limitation is exposed explicitly — never fabricated as a complete result.

31.3 Observability — Technical vs Governance Indicators

Table 13. Memory observability — technical and governance indicators tracked separately; operational metrics are never confused with Knowledge quality (front matter invariant, item 47).

Category	Indicators
Technical	Retrieval latency, search lag, graph refresh lag, vector index health, archive retrieval time
Governance	Reconciliation failures, snapshot integrity, Knowledge review backlog, retracted Knowledge with active dependencies

31.4 Performance Boundary

Temporal integrity, authorization, provenance, historical reconstruction and Knowledge governance are never sacrificed for retrieval speed. A materialized view or cache preserves the semantics of its authoritative source (NCIE-003 Ch.32 §32.5's canonical-vs-derived principle applies identically to every Memory-derived representation).

31.3 Agent Context Observability (v1.2)

Agent Context telemetry is observable using this chapter's existing performance/observability architecture, subject to current authorization, purpose, classification and minimum-necessary scope for telemetry privacy — Dynamic Agent/VPF capability must not broaden telemetry access or protected-identity exposure. Context leakage monitoring and Context quality observability extend the existing signal categories without collapsing into one generic health metric.

Traceability: NCIE-004 Ch.31, Ch.35; NCIE-003 Ch.32.

Resilience, Backup, Recovery & Post-Restore Reconciliation

Recover Memory/Knowledge without reactivating superseded state or rewriting original knowledge time.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-004 Ch.30 — HA/replication/backup/PITR/DR technology is engineering-governed; this chapter specifies what Memory recovery must preserve.

32.1 Failure and Degraded Modes

Table 14. Memory failure/degraded-mode behavior by dependency — infrastructure failure is never hidden by fabricated context.

Dependency Lost	Behavior
Authoritative domain state unavailable	Reconciliation returns CURRENT STATE UNVERIFIED; historical Memory is never presented as verified current truth
Search unavailable, canonical stores healthy	Alternative governed retrieval used where available; search-index loss is never treated as Memory loss
Vector index unavailable	Degrades to authorized keyword/structured/graph retrieval; ARGUS never implies no Knowledge exists merely because semantic retrieval is unavailable
Graph unavailable	Graph-derived analytics degrade; canonical Memory/Knowledge objects remain authoritative in their source stores
Reconciliation fails	Returns CURRENT STATE UNVERIFIED rather than a false verified result
Knowledge Service unavailable	Retrieval of Approved Knowledge degrades gracefully; ARGUS states the limitation rather than fabricating remembered context
Snapshot recovery fails	Failure is surfaced explicitly; a partial/corrupt snapshot is never silently treated as complete

32.2 Backup / Recovery / Post-Restore Reconciliation Flow

Figure 13. Backup / Recovery / Post-Restore Reconciliation Flow — restoration alone is never treated as sufficient.

01	Backup captures Decision-Time Snapshots, Knowledge governance state, Memory provenance, classification, legal holds and dependency relationships (highest recovery priority)
02	Restoration from backup
03	Post-restore reconciliation: Knowledge Status, Retention State, Authorization State, Rules and current domain references checked against current authoritative governance
04	Knowledge-time values preserved as originally recorded (KNOWN AT ≠ RESTORED AT)
05	Superseded/retracted Knowledge is not reactivated merely because an old backup contained it

32.3 Recovery Priority and Integrity

- High recovery priority: Decision-Time Snapshots, Knowledge governance state, Memory provenance, classification, legal holds, dependency relationships.
- Recovery preserves temporal semantics, Knowledge versions, supersession, retraction, Decision-Time Snapshots, graph provenance, classification and audit relationships — an old backup never reactivates obsolete Knowledge.
- Recovery preserves the original KNOWN AT (knowledge-time) value; RESTORED AT never substitutes for it.

32.3 Agent/VPF Recovery Reconciliation (v1.2)

RECOVERY ≠ REAUTHORIZATION: a restored backup may contain historical authorization, Agent eligibility, Tool/model eligibility or VPF Profile state; none becomes current merely because the backup was restored. Post-restore reconciliation applies this chapter's existing discipline to Agent Context (expired Context does not reactivate), Agent eligibility (current Registry state prevails over backup-era state), VPF state (obsolete/suspended Profiles do not become current) and vector/search indexes (current deletion/restriction/classification/authorization state is respected before ordinary retrieval resumes). Historical Validation Artifacts and Decision-Time state are preserved through recovery unchanged. Where Agent/VPF Context integration is disabled or degraded, unrelated deterministic Memory capability remains available where its independent dependencies are healthy.

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm RTO/RPO targets for Memory specifically, or confirm inheritance of NCIE-004 Ch.30's proposed tier defaults without a separate NCIE-006 decision.

Traceability: NCIE-004 Ch.30.

Legacy Memory Migration & Historical Identity Resolution

Import historical records with source/transformation provenance and explicit ambiguous-identity handling.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.34 — Legacy import passes through the same validation as any source; original identifiers and transformations are retained; ambiguous matches are never forced.

33.1 Migration Requirements

Imported legacy records are never treated as validated Institutional Knowledge merely by virtue of import. Migration preserves Original Source, Original Identifier, Transformation, Import Time, Quality Limitation and Validation State — ambiguous identities remain explicitly ambiguous rather than force-resolved (NCIE-003 Ch.34 §34.1's rule applies identically to Memory/Knowledge migration).

33.2 Technology Migration Neutrality

Changing database, search engine, vector store, graph platform, model provider or hosting environment never changes canonical identity, provenance, authority, temporal meaning, classification or Knowledge governance state (NCIE-003 Ch.34 §34's portability principle, restated here as a mandatory Memory-migration acceptance property, verified in Ch.34 of this document).

33.3 Agent/VPF Migration Integrity (v1.2)

Provider/framework changes must not destroy NCIE-controlled Memory/Context semantics. PROVIDER MEMORY ≠ NCIE INSTITUTIONAL MEMORY: provider-native conversation/Agent/thread/vector memory or session persistence is treated as implementation/runtime capability, not NCIE Institutional Memory, unless explicitly governed and mapped; provider-native retrieval must not bypass NCIE authorization, purpose, classification, Context scope or Decision-Time controls, and provider-native embeddings remain subject to this document's classification, deletion, restriction, residency, authorization and source-provenance requirements. MIGRATED / IMPORTED ≠ APPROVED / ELIGIBLE: migration/export preserves enough metadata to reconstruct governed relationships among Memory Objects, Context Envelopes, Agent Runs, Evidence, Knowledge, Validated AI Results, VPF Validation Artifacts and Human Decisions; imported Memory is checked for identity, version, integrity, temporal meaning, provenance and classification before ordinary use, and where historical Agent/VPF/Context metadata cannot be migrated, the limitation is identified explicitly rather than fabricated. Imported historical state does not establish current authorization or eligibility.

Human Review Focus

- **[SOURCE/WORKFLOW DISCOVERY REQUIRED]** Inventory legacy memory/knowledge repositories (if any) requiring migration and their current owners — no existing NCA repository is assumed.

Traceability: NCIE-003 Ch.34.

Memory & Context Testing, Verification & Acceptance

Test memory classes, bitemporality, reconciliation, authorization, knowledge lifecycle, deletion, recovery and model replacement.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 Ch.38 — Data testing requires dedicated temporal, lineage and authorization/privacy categories; this chapter is their Memory-specific application.

34.1 Required Test Categories

At minimum: Memory-class isolation; private/shared context isolation; permission change; Decision-Time reconstruction; late evidence; current-state reconciliation; Knowledge promotion/challenge/supersession/retraction; dependency impact; semantic search authorization; graph traversal authorization; protected identity; voice transcript handling; retention; legal hold; derived deletion; backup restore; model replacement; and staff continuity.

34.2 Adversarial Memory Tests

Adversarial scenarios deliberately attempt to make ARGUS:

- Treat Historical Memory as Current Truth.
- Treat Conversation as Evidence.
- Treat its own prior output as Approved Knowledge.
- Use Retracted Knowledge as current.
- Leak restricted Memory through search, or restricted graph topology.
- Follow instructions embedded in historical Memory.
- Promote its own Knowledge Candidate.
- Infer identity from voice.
- Treat a historical authorization grant recorded in Memory as still valid today.
- Automatically replay a prior Decision as authorization for a new consequential action.

All must fail safely (pass criterion: zero successful instances in each scenario).

34.3 Mandatory Acceptance Tests

Table 15. Mandatory Memory & Context acceptance tests.

Test	Demonstrates
Model replacement test	Replacing the ARGUS model/provider does not erase Institutional Memory.
Staff-continuity test	Institutional Memory survives staff departure, role change, shift change, vendor change, system migration and model replacement.
Decision reconstruction test	A historical Decision's Evidence/Rules/Hypotheses/Contradictions/Gaps/ARGUS Contribution/Human Decision Context reconstruct without contamination by later Evidence.
Current-vs-historical test	Both HISTORICAL STATE and CURRENT AUTHORITATIVE STATE are shown for a changed object without conflation.
Knowledge retraction test	Current retrieval stops presenting retracted Knowledge as active; historical trace remains; dependent active matters are flagged; derived representations update; audit is preserved.
Remembered-authorization test	A historical grant of authorization recorded in Memory (including within a Decision-Time Snapshot) never substitutes for a current authorization check (Ch.25 §25.3); a revoked, expired or role-changed permission is never reinstated by recollection.
Decision-replay test	A historical Decision (Ch.11 §11.4) is never automatically replayed as authorization for a new blocking, tracking, regulatory or other consequential action absent a fresh current human Decision.

34.3 Agent/VPF Memory & Context Test Suite (v1.2)

Extend this chapter's existing test catalogue with the minimum Agent/VPF Memory/Context tests necessary to prove the amended architecture — not a duplicate of NCIE-004/007's complete Agent/VPF test catalogues, focused on Memory/Context consequences: Agent Context isolation across Agents/users/Rooms/Investigations; minimum-necessary Context assembly and Context denial outside scope; current-authorization and remembered-authorization tests; Agent expiry (retrieval stops, runtime state cleaned, required audit/provenance/Decision-Time records survive); Agent Result persistence distinguishable from Evidence/Knowledge/Judgment/Decision; VPF-PASS-does-not-equal-Knowledge tests; Evidence withdrawal and Knowledge retraction dependency-impact tests; Decision-Time reconstruction using historical Agent/VPF/Tool/Context state exclusively; current-state reconciliation against stale Memory; semantic-similarity and deleted-embedding disclosure tests; prompt-injection Context tests; voice Context and spoken-disclosure tests; human-modification-after-PASS tests; multi-Agent and child-Agent Context isolation; cross-task cache authorization tests; and recovery tests (authorization, Agent Context, Agent eligibility, VPF state, vector-index state). TEST SPECIFIED ≠ TEST EXECUTED — new tests are not marked PASS merely because they are specified in this document.

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm required test environments and masked test data provisioning, consistent with NCIE-003 Ch.29 §29.3's convention.

Traceability: NCIE-003 Ch.38.

Human Review Register, Open Decisions & Proposed Defaults

Consolidate every unresolved Memory/Context decision with category, owner and blocking status.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 — — Consolidates every Human Review Focus item from Chapters 1-34 into one authoritative register.

35.1 Purpose

Consolidates every Human Review Focus item raised in Chapters 1-34, extracted directly from the chapter review blocks so this register cannot drift from its sources. Where an item inherits an already-open NCIE-004/005 decision rather than introducing a new NCIE-006 question, it is noted as such in the Issue text rather than duplicated under an unrelated ID (handover items 164-165).

35.2 Summary

Table 16. Human Review Register summary counts.

Metric	Count
Total items	20
Resolved by Upstream Baseline	1
Proposed Design Default	2
Institutional Confirmation Required	9
Source/Workflow Discovery Required	2
Legal/Policy Confirmation Required	4
Security/Sovereignty Confirmation Required	2
Blocking	17
Non-Blocking	3

35.3 Complete Register

Table 17. Complete Human Review Register (Chapters 1-34).

ID	Ch.	Issue	Category	Blocking?
HR6-1-1	Ch.1	Confirm the acceptance authority for NCIE-006 as a whole and for individual chapter sign-off.	INSTITUTIONAL CONFIRMATION REQUIRED	BLOCKING
HR6-2-1	Ch.2	All five epistemic boundaries in §2.2 are fixed by NCIE-002 Ch.20 and NCIE-003 Ch.22; no institutional decision is required to adopt them.	RESOLVED BY UPSTREAM BASELINE	NON-BLOCKING
HR6-3-1	Ch.3	Confirm retention, access or ownership rules that differ by Memory class beyond what each owning chapter proposes.	INSTITUTIONAL CONFIRMATION REQUIRED	BLOCKING

ID	Ch.	Issue	Category	Blocking?
HR6-4-1	Ch.4	The Context Envelope field set in §4.1 is proposed as complete for launch; additional fields may be added additively without a breaking change (NCIE-003 Ch.2 §2.8 pattern).	PROPOSED DESIGN DEFAULT	NON-BLOCKING
HR6-4-2	Ch.4	Confirm Agent Context / Agent Run Context retention period specifically, distinct from ordinary Task/Run lifecycle expiry (Ch.4 §4.4, Ch.28 §28.4).	INSTITUTIONAL CONFIRMATION REQUIRED	BLOCKING
HR6-5-1	Ch.5	Confirm Conversation Memory retention period — no universal period is assumed; classified per Ch.28's class-specific retention requirement.	LEGAL/POLICY CONFIRMATION REQUIRED	BLOCKING
HR6-8-1	Ch.8	Confirm audio retention policy specifics with NCA (inherits NCIE-005's open item on audio/transcript retention as a separate decision, per §0.2's no-duplication rule — see NCIE-005 Ch.29 §29 review).	SOURCE/WORKFLOW DISCOVERY REQUIRED	BLOCKING
HR6-12-1	Ch.12	UTC as the canonical knowledge-time storage basis is proposed, consistent with NCIE-003 Ch.5 §5.8's platform-wide default.	PROPOSED DESIGN DEFAULT	NON-BLOCKING
HR6-18-1	Ch.18	Confirm whether separation of duties between Candidate Contributor, Reviewer and Approver is institutionally required, or whether one authorized role may perform more than one step.	INSTITUTIONAL CONFIRMATION REQUIRED	BLOCKING
HR6-18-2	Ch.18	Confirm whether Agent/Validated-AI-Result-originated Knowledge Candidates require additional review steps beyond the existing Candidate Contributor/Reviewer/Approver separation (HR6-18-1).	INSTITUTIONAL CONFIRMATION REQUIRED	BLOCKING
HR6-19-1	Ch.19	Confirm review-due default behavior (ACTIVE-REVIEW-DUE vs SUSPENDED) per Knowledge class where NCA policy is not otherwise specified.	INSTITUTIONAL CONFIRMATION REQUIRED	BLOCKING
HR6-22-1	Ch.22	Confirm operational ownership for Agent/VPF dependency impact-analysis triggers (VPF Profile, Agent Definition, Model, Tool and Human Review changes) introduced by this amendment.	INSTITUTIONAL CONFIRMATION REQUIRED	BLOCKING
HR6-26-1	Ch.26	Confirm audio/transcript retention periods and applicable privacy policy — this is the same open question already raised at NCIE-005 Ch.29 §29-review; not duplicated here as a new decision.	LEGAL/POLICY CONFIRMATION REQUIRED	BLOCKING
HR6-26-2	Ch.26	Confirm cross-border processing boundaries for any voice/transcript content that could leave an approved jurisdiction via cloud STT/TTS processing (NCIE-004 Ch.18's provider eligibility applies).	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED	BLOCKING

ID	Ch.	Issue	Category	Blocking?
HR6-28-1	Ch.28	Confirm retention periods per Memory class (Ch.3 §3.2) with NCA Legal/Records Management; none is assumed.	LEGAL/POLICY CONFIRMATION REQUIRED	BLOCKING
HR6-28-2	Ch.28	Confirm how personal deletion rights interact with institutional records obligations where they may conflict — no universal precedence is assumed.	LEGAL/POLICY CONFIRMATION REQUIRED	BLOCKING
HR6-32-1	Ch.32	Confirm RTO/RPO targets for Memory specifically, or confirm inheritance of NCIE-004 Ch.30's proposed tier defaults without a separate NCIE-006 decision.	INSTITUTIONAL CONFIRMATION REQUIRED	BLOCKING
HR6-33-1	Ch.33	Inventory legacy memory/knowledge repositories (if any) requiring migration and their current owners — no existing NCA repository is assumed.	SOURCE/WORKFLOW DISCOVERY REQUIRED	BLOCKING
HR6-33-2	Ch.33	Confirm the governance/mapping requirement before any provider-native Agent/session memory may be treated as NCIE Institutional Memory (PROVIDER MEMORY ≠ NCIE INSTITUTIONAL MEMORY, Ch.33 §33.3).	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED	BLOCKING
HR6-34-1	Ch.34	Confirm required test environments and masked test data provisioning, consistent with NCIE-003 Ch.29 §29.3's convention.	INSTITUTIONAL CONFIRMATION REQUIRED	BLOCKING

35.4 Mechanical Verification

Per handover items 49/110, this register's totals are verified programmatically, not by inspection: the production build independently re-extracts every ('review', ...) block across Chapters 1-34 and asserts the result matches this chapter's own extraction (§35.2-§35.3) exactly, by category and by blocking status; a mismatch fails the build rather than reaching this document. The current build confirms an exact match — 16 items (1 Resolved / 2 Proposed / 6 Institutional / 2 Workflow / 4 Legal / 1 Security), 13 Blocking / 3 Non-Blocking — confirming no item was silently dropped or miscounted. This figure and §35.2's summary table are generated from the same computed extraction at build time, so they cannot diverge from one another.

35.5 Closure and Blocking Scope

An item closes only when its required confirmation is actually supplied and recorded — never converted into an invented fact merely to appear resolved. A Blocking item blocks only the specific chapter/capability that genuinely depends on it — retention period, storage technology or audio retention being unresolved does not block the entire 36-chapter specification; technology-neutral, policy-hook design proceeds where possible (handover item 166).

35.6 v1.1 → v1.2 Reconciliation Note

This edition adds 4 new Human Review items (HR6-4-2, HR6-18-2, HR6-22-1, HR6-33-2) introduced by the VPF & Dynamic Agent Context Surgical Consistency Amendment. No existing v1.1 item was removed or silently resolved; none is conclusively resolved by the current approved upstream authorities (all remain genuine open institutional/security questions). Totals recomputed mechanically above from final register content: 20 total (1 Resolved / 2 Proposed / 9 Institutional / 2 Workflow / 4 Legal / 2 Security), 17 Blocking / 3 Non-Blocking. Reconciles as 16 (v1.1) + 4 added – 0 removed = 20.

Traceability: Consolidates Chapters 1-34; individual chapters trace to NCIE-002/003/004/005 as cited therein.

Requirements Traceability, Acceptance & Handover Gate

Crosswalk NCIE-006 to NCIE-002-005 and define controlled acceptance and handover to NCIE-007.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 — — Closes the authority chain: NCIE-001 requirements → NCIE-002 architecture → NCIE-003 canonical data → NCIE-004 engineering → NCIE-005 ARGUS behavior → NCIE-006 Memory/Context.

36.1 Traceability Matrix

Table 18. NCIE-006 chapter-to-upstream traceability matrix.

Ch.	Title	Primary Upstream Dependency
1	Mandate, Authority Chain & Specification Boundary	NCIE-002 Ch.20; NCIE-003 Ch.22; (v1.2) NCIE-004 v1.2; NCIE-005 v1.3; NCIE-007 v1.1
2	Memory Doctrine, Invariants & Epistemic Boundaries	NCIE-002 Ch.20 §20.2, §20.12; NCIE-003 Ch.22; (v1.2) NCIE-003 v0.4 Ch.3 §3.9; NCIE-007 v1.1
3	Memory Class Taxonomy	NCIE-002 Ch.20 §20.2; (v1.2) NCIE-007 v1.1
4	Context Taxonomy & Context Envelope	NCIE-002 Ch.19 §19.6; (v1.2) NCIE-003 v0.4 Ch.3 §3.9; NCIE-007 v1.1
5	Conversation Memory	NCIE-005 Ch.5, Ch.11; (v1.2) NCIE-005 v1.3; NCIE-007 v1.1
6	User Preference Memory	NCIE-005 Ch.28
7	Workspace & Session Continuity Memory	NCIE-002 Ch.3 §3.1
8	Collaborative Room Memory	NCIE-002 Ch.20 §20.5, §20.10; NCIE-003 Ch.21
9	Investigation Memory	NCIE-003 Ch.15 §15.3
10	Situation Memory	NCIE-003 Ch.18
11	Decision Memory & Decision-Time Context	NCIE-003 Ch.16, Ch.25, Ch.26; (v1.2) NCIE-003 v0.4; NCIE-007 v1.1
12	Historical Memory & Bitemporal Reconstruction	NCIE-003 Ch.5, Ch.25
13	Current-State Reconciliation	NCIE-002 Ch.20 §20.10; NCIE-005 Ch.23
14	Context Assembly & Minimum-Necessary Retrieval	NCIE-002 Ch.24 §24.3; (v1.2) NCIE-004 v1.2 Ch.20; NCIE-007 v1.1
15	Context Propagation Across Services & Workflows	NCIE-002 Ch.4 §4.4; (v1.2) NCIE-007 v1.1
16	Private, Shared & Institutional Context Boundaries	NCIE-002 Ch.20 §20.9; NCIE-003 Ch.4
17	ARGUS Memory Retrieval Interface	NCIE-005 Ch.11, Ch.23; (v1.2) NCIE-005 v1.3; NCIE-007 v1.1

Ch.	Title	Primary Upstream Dependency
18	Institutional Knowledge Candidate Lifecycle	NCIE-003 Ch.22 §22.4; NCIE-005 Ch.24; (v1.2) NCIE-003 v0.4 Ch.3 §3.9; NCIE-005 v1.3; NCIE-007 v1.1
19	Knowledge Versioning, Applicability & Review	NCIE-003 Ch.22 §22.3
20	Knowledge Challenge, Contradiction & Impact Analysis	NCIE-005 Ch.24; NCIE-003 Ch.22
21	Knowledge Graph Context & Memory Relationships	NCIE-003 Ch.19 §19.2, §19.7
22	Dependency Graph & Downstream Impact Tracking	NCIE-003 Ch.25 §25.4; (v1.2) NCIE-003 v0.4; NCIE-007 v1.1
23	Memory Search, Retrieval Modes & Ranking	NCIE-003 Ch.4 §4.89-4.93
24	Semantic Memory & Vector Retrieval Boundaries	NCIE-002 Ch.26 §26.3; NCIE-004 Ch.16
25	Memory Security, Authorization & Zero-Trust Controls	NCIE-002 Ch.4; (v1.2) NCIE-004 v1.2 Ch.23; NCIE-007 v1.1
26	Sensitive Memory, Protected Identity & Voice Context	NCIE-003 Ch.14 §14.6, Ch.36; NCIE-005 Ch.29; (v1.2) NCIE-005 v1.3 Ch.29; NCIE-007 v1.1
27	Memory Provenance, Attribution & Audit	NCIE-002 Ch.24 §24.4; NCIE-005 Ch.27; (v1.2) NCIE-007 v1.1
28	Retention, Legal Hold, Archival & Secure Disposition	NCIE-003 Ch.31; (v1.2) NCIE-003 v0.4 Ch.3 §3.9
29	Memory Storage Architecture & Canonical/Derived Stores	NCIE-004 Ch.11-17, Ch.33; (v1.2) NCIE-004 v1.2 Ch.33 §33.4
30	Memory APIs, Events & Service Contracts	NCIE-004 Ch.8-9; NCIE-003 Ch.28; (v1.2) NCIE-007 v1.1
31	Scalability, Tiering, Performance & Observability	NCIE-004 Ch.31, Ch.35; NCIE-003 Ch.32; (v1.2) NCIE-004 v1.2
32	Resilience, Backup, Recovery & Post-Restore Reconciliation	NCIE-004 Ch.30; (v1.2) NCIE-004 v1.2 Ch.30
33	Legacy Memory Migration & Historical Identity Resolution	NCIE-003 Ch.34; (v1.2) NCIE-003 v0.4
34	Memory & Context Testing, Verification & Acceptance	NCIE-003 Ch.38; (v1.2) NCIE-004 v1.2; NCIE-007 v1.1
35	Human Review Register & Open Decisions	Consolidates Ch.1-34; (v1.2) Reconciles the Ch.2,4,5-10,11-13,14,15,17,18,22,25-33 Agent/VPF Human Review items added by this amendment

Ch.	Title	Primary Upstream Dependency
36	Traceability, Acceptance & Handover	Closes NCIE-001→002→003→004→005→006 authority chain; (v1.2) NCIE-001 v1.3; NCIE-002 v0.5; NCIE-003 v0.4; NCIE-004 v1.2; NCIE-005 v1.3; NCIE-007 v1.1

36.2 Traceability Mechanical Verification

Per handover item 111, every one of the 36 chapters above carries at least one upstream citation — verified by inspection of §36.1. No ORPHAN NCIE-006 REQUIREMENT (a chapter with no upstream justification) or UNREALIZED UPSTREAM REQUIREMENT (an NCIE-002-005 Memory/Context mandate with no corresponding NCIE-006 chapter) was found in this edition.

36.3 Acceptance Checklist

- All ten Governing Invariants (front matter §0.3) are preserved without exception across all 36 chapters.
- The nine Memory classes (Ch.3 §3.2) each have distinct purpose, authority, retention and reconciliation treatment — none collapsed into a generic store or lifecycle.
- Decision-Time reconstruction (Ch.11-12) excludes later Evidence in every worked example.
- Institutional Knowledge promotion/challenge/supersession/retraction (Ch.18-20) requires human action at every authority-changing step; ARGUS never approves.
- Voice-derived Memory (Ch.8 §8.2, Ch.26) preserves the NCIE-005 voice invariants without modification.
- No chapter fully specifies NCIE-007/009/010/011/014/016; each cross-references the downstream owner (Ch.1 §1.2 boundary table).
- No technology named from NCIE-004 is asserted as more approved here than its NCIE-004 Technology Registry status.
- Chapter 35's Human Review Register is complete and mechanically verified against its source chapters (§35.4).
- Chapter 36's traceability matrix is mechanically verified complete (§36.2).

36.4 Handover

On acceptance, this Memory & Context architecture hands forward to NCIE-007 (AI, Agent & Model Orchestration), NCIE-009 (Security/IAM), NCIE-010 (Evidence, Provenance & Audit), NCIE-011 (API/Integration), NCIE-014 (Database & Storage Engineering) and NCIE-016 (Testing) — each of which implements this document's Memory/Context requirements without re-deciding the Governing Invariants or Memory Class boundaries this document fixes.

36.5 Current Authority Chain (v1.2)

NCIE-001 v1.3 — APPROVED. NCIE-002 v0.5 — APPROVED. NCIE-003 v0.4 — APPROVED. NCIE-004 v1.2 — APPROVED. NCIE-005 v1.3 — APPROVED. NCIE-007 v1.1 — APPROVED. NCIE-006 does not treat its own v1.1 citations to NCIE-001 v1.2, NCIE-002 v0.4, NCIE-003 v0.3, NCIE-004 v1.1 or NCIE-005 v1.2 as current authority; those remain preserved only as historical baseline references.

36.6 Agent/VPF Memory & Context Acceptance (v1.2)

In addition to §36.3's existing checklist, this edition confirms: NCIE-006 owns Memory and Context semantics only — it does not become Agent Factory, Agent Registry, Model Registry, Tool Registry, VPF, institutional authorization authority or physical storage specification, and no chapter absorbs those downstream/upstream specifications. Dynamic Agents receive bounded task/run-specific Context through governed Context assembly and retrieval (Ch.4, Ch.14). Agent Runtime State does not become Institutional Memory by default; Agent Context does not create Agent authority; Context propagation does not propagate permission; Agent-to-Agent delegation does not automatically propagate full parent Context (Ch.15). Decision-Time reconstruction preserves the historical Agent/VPF/Context state available at the cutoff without later contamination (Ch.11-13). MEMORY ≠ CURRENT TRUTH is preserved for all consequential current-state resolution. The Evidence and Knowledge boundaries are preserved: Agent analysis of Evidence is not Evidence itself, and a Validated AI Result does not automatically become Institutional Knowledge (Ch.2, Ch.18). TEST SPECIFIED ≠ TEST EXECUTED is preserved for all new Ch.34 tests.

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm NCIE-006 approvers and their scope of authority.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm which of Ch.35's Blocking items must close before NCIE-007/009/010/011/014/016 may begin detailed implementation versus which may proceed provisionally.
- **Confirm which of Chapter 35's new Blocking items (HR6-4-2 Agent Context retention, HR6-18-2 Knowledge Candidate review steps, HR6-22-1 dependency-impact ownership, HR6-33-2 provider-native memory governance) must close before NCIE-007/009/010/011/014/016 may begin detailed implementation versus which may proceed provisionally.**

Traceability: Closes the NCIE-001 → NCIE-002 → NCIE-003 → NCIE-004 → NCIE-005 → NCIE-006 authority chain (front matter §0.1).